

A Machine-Checked Finite Reduction for Erdős Problem #287 and a Source-Explicit Large- M Criterion

Clean Public Review + Self-Contained Gate Theory Manual
Version 16.8 – Public Review R9 (1 September 2026 source reconstruction)

Research draft with Lean/Aristotle-assisted formal verification

1 September 2026

Status – 1 September 2026 source reconstruction. Erdős Problem #287 remains open. The unconditional Lean exclusion up to 4×10^9 is unchanged. The newest source archaeology has now recovered the literal Ford–Maynard G/H source and the true labelled seven-prime-vector master index. On the corrected strictly smooth balanced-seven cell the paper-level source calculation proves $H_*(P) = -20$, including the strict Ford threshold and the required factor-8 three-prime bound. The exact vector-to-integer pushforward carries the physical fibre coefficient $\Omega_C^\#(n)$; it is therefore not silently identified with the older unweighted integer source. The next controlling source/compiler nodes are:

BALANCED7-SP2-STRICTCELL-PHYSICAL-BRIDGE45
then
FORD723-GENERATED-PACKET-CENSUS-AND-REPLACEMENT45.

The generated Ford packet census, exact physical Perron census, full-source local analytic kernel, and final large- M /public-predicate bridge remain open.

A supplied append-only Aristotle master-source-frontier bank reports **lake build**: 8330 jobs, zero errors, with only ordinary Mathlib foundations in its axiom audit. That formal bank is used here as a verification checkpoint, while its generic master-source/physical-dictionary interfaces remain explicitly uninhabited unless the newer physical source reconstruction supplies them. No theorem named **erdos287** and no **Erdos287ClosureInputs** inhabitant are claimed.

Abstract

Erdős Problem #287 asks whether every representation

$$1 = \frac{1}{n_1} + \cdots + \frac{1}{n_k}, \quad 1 < n_1 < \cdots < n_k,$$

contains a consecutive gap of size at least 3. The strongest unconditional theorem in this project remains the machine-checked exclusion of exact counterexamples with maximum denominator at most 4×10^9 ; the large- M problem remains open.

This R9 synchronization re-anchors the later analytic packet work to the literal physical source. Ford–Maynard’s G/H factorisation has been recovered at paper level. On the strictly smooth $k = 0, J = \emptyset$ cell it collapses pointwise to the truncated Möbius weight. For seven pairwise distinct primes $p_i \in [Y, 2Y)$ with $Y \geq 256$, the strict threshold and depth-three divisor inequalities give

$$\boxed{H_*(P) = 1 - 7 + 21 - 35 = -20}, \quad P = \prod_{i=1}^7 p_i.$$

The genuine earliest master index is the labelled prime vector, not the downstream $C0$ /transverse/ b -diagonal coordinates. Pushing the vector source to integers introduces

the exact fibre weight $\Omega_C^\#(n)$, so the physical integer source is weighted and is not automatically the older unweighted P_{sm} .

A supplied append-only Aristotle master-source-frontier bank reports 8330 build jobs and zero errors and formally banks the source-realisation firewalls, proof-local Ω scaffolding, typed packet compiler, Perron counterguards, and abstract reciprocal finite Fourier kernels. Physical realisation interfaces remain uninhabited where appropriate. The next research task is the literal Ford-(7.23) generated-packet census and replacement by source-specific local owners, followed by exact Perron/source reassembly. No full-source local analytic closure, no eventual WindowPairSupply, and no unconditional solution of Erdős #287 are claimed.

Status at a glance

Item	Status	Meaning
Finite $M \leq 4 \times 10^9$ exclusion	KERNEL-PROVED / UNCHANGED	Strongest unconditional theorem.
Supplied master-source-frontier Aristotle bank	KERNEL-CHECKED BUNDLE	8330 jobs, zero errors; physical interfaces remain explicit.
Ford G/H source	SOURCE FOUND	Authoritative source recovered at paper level.
Balanced-seven strict cell / $H_*(P) = -20$	PAPER/RESEARCH PASS	Strict threshold and factor-8 repair included.
Labelled prime-vector master source	PAPER/RESEARCH PASS	Correct earliest physical index recovered.
Weighted integer pushforward	PAPER/RESEARCH PASS	Carries $\Omega_C^\#(n)$; unweighted equality is not automatic.
BALANCED7-SP2-STRICTCELL-PHYSICAL-BRIDGE45	FIRST PHYSICAL BRIDGE / OPEN	Bind literal Ford/source fields into the public packet.
FORD723-GENERATED-PACKET-CENSUS-AND-REPLACEMENT45	NEXT SOURCE/COMPILER FRONTIER / OPEN	Exhaust generated branches and assign legal owners.
MASTER-SOURCE-T0-TYPED-PERRON-PACKETS45	OPEN	Exact generated packet/contour census incomplete.
FULL-SOURCE-LOCAL-ANALYTIC-KERNEL45	OPEN	No 100% physical source coverage claim.
WindowPairSupply / final large- M bridge	OPEN	End-to-end public-predicate closure not supplied.
Erdős Problem #287	OPEN	No unconditional solution claim.

Status vocabulary. KERNEL-PROVED is reserved for kernel-checked statements. PAPER/RESEARCH PASS means a source/mathematical argument has survived the current research audit but has not yet been promoted to the full physical Lean theorem. OPEN means precisely that the stated source, compiler, analytic, or global bridge is not discharged.

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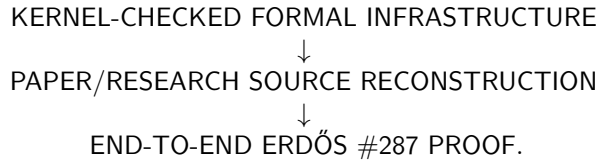
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1 1 September 2026 source reconstruction and formal-frontier update

This section supersedes the 31-August C0/transverse checkpoint as the controlling status summary. It does not erase the earlier analytic work: it re-anchors that work to the literal Ford–Maynard source and separates three levels throughout:



The third level remains OPEN.

1.1 Kernel-checked master-source-frontier bank

A supplied append-only Aristotle pass reports a master-source-frontier formal bank with `lake build`: 8330 jobs, zero errors, and no new-module warnings. Its axiom audit reports only ordinary Mathlib foundations (principally `propext`, `Classical.choice`, and `Quot.sound`); it reports no `sorryAx`, custom axiom, `unsafe`, `native_decide`, or `implemented_by` code. At the time of this R9 synchronization those newest formal files are treated as a supplied formal checkpoint rather than silently identified with the older public-main tree.

The formal bank safely proves infrastructure including: a source-dictionary versus physical-realisation firewall; an abstract proof-local dyadic Ω partition with exact reconstruction and bounded overlap; a typed packet decomposition with an explicit *global* nuclear-mass field; counterguards showing that per-contour or per-packet bounds do not imply a global total without cardinality control; a deterministic packet compiler; and an abstract $\Delta \times q_a$ reciprocal finite-Fourier kernel. The actual physical master-source realisation, physical Ω insertion, physical C0/transverse/ *b*-diagonal dictionaries, full source coverage, and eventual `WindowPairSupply` remain uninhabited/open at that formal checkpoint.

1.2 Ford source recovered at paper/research level

The authoritative source is Ford–Maynard, *On the theory of prime-producing sieves*, <https://arxiv.org/abs/2407.14368>. Their source architecture includes

$$G(m; n) = \mathbf{1}_{m \leq n^\gamma} \lambda(m_1) g(\mathbf{v}(m_2; n)), \quad m = m_1 m_2,$$

with threshold factorisation

$$P^+(m_1) < n^\sigma \leq P^-(m_2), \quad H(n) = \sum_{d|n} G(d; n).$$

For $\lambda = \mu$ on the strictly smooth cell $P^+(n) < n^\sigma$, the threshold factorisation forces $m_1 = d$, $m_2 = 1$. Since the fixed certificate has $g_*(\emptyset) = 1$, one obtains pointwise

$$H_*(n) = \sum_{\substack{d|n \\ d \leq n^\gamma}} \mu(d) = \text{truncMobius}(n, \lfloor n^\gamma \rfloor).$$

The strict inequality is load-bearing: the generic non-strict condition $P^+(n) \leq n^\sigma$ is not promoted to the same statement.

1.3 Balanced-seven strict physical cell

Let

$$\nu_0 = 0.16623, \quad \sigma_* = \nu_0 - 2\varepsilon_*, \quad \gamma_* = \frac{1}{2} - \varepsilon_*,$$

with $0 \leq \varepsilon_* \leq \nu_0/100$. On the intended cell, take $Y \geq 256$ and seven pairwise distinct primes $Y \leq p_i < 2Y$, and put $P = \prod_{i=1}^7 p_i$. The repaired source calculation proves

$$P^+(P) < P^{\sigma_*}.$$

For a divisor $d_S = \prod_{i \in S} p_i$, the correct three-prime bound is $d_S < (2Y)^3 = 8Y^3$, not Y^3 . Together with the four-or-more-prime lower bound this gives

$$d_S \leq P^{\gamma_*} \iff |S| \leq 3.$$

Consequently the fixed-certificate weight on this physical cell is

$$H_*(P) = \sum_{j=0}^3 (-1)^j \binom{7}{j} = 1 - 7 + 21 - 35 = -20.$$

This is classified here as PAPER/RESEARCH-LEVEL PASS. The corresponding physical Lean bridge still has to bind the dyadic prime-box support, pairwise distinctness, literal Ford H_s , and the cut $\lfloor P^{\gamma_*} \rfloor$ to the generic public packet type.

1.4 Correct earliest master source

The genuine earliest summed index is not the later tuple of C0/transverse/ b -diagonal coordinates. It is the labelled seven-prime vector

$$\boxed{\xi = \mathbf{p} \in C.\text{cell}}, \quad P(\mathbf{p}) = \prod_{i=1}^7 p_i.$$

The labelled physical source is

$$S_{\text{vec}} := \sum_{\mathbf{p} \in C.\text{cell}} C.\Omega(\mathbf{p}) W\left(\frac{P(\mathbf{p})}{X}\right) \times \left[\Lambda(2P(\mathbf{p}) - 1) + \Lambda(2P(\mathbf{p}) + 1) - 4C.B(P(\mathbf{p})) \right] H_s(P(\mathbf{p})).$$

Coordinates such as q, e, A_0, C_0, Δ and shared-gcd data are descendants of this source and are not promoted to fields of the earliest master index.

1.5 Exact weighted integer pushforward

Define the physical product image $\mathcal{N}_C$ and fibre coefficient

$$\Omega_C^\#(n) := \sum_{\substack{\mathbf{p} \in C.\text{cell} \\ P(\mathbf{p})=n}} C.\Omega(\mathbf{p}).$$

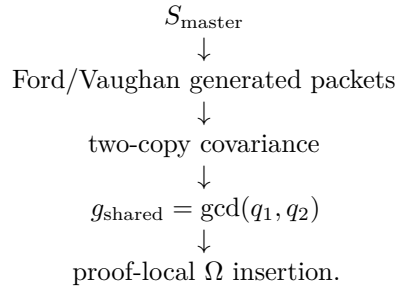
Then finite fibre grouping gives the exact identity

$$S_{\text{vec}} = \sum_{n \in \mathcal{N}_C} \Omega_C^\#(n) W(n/X) \times [\Lambda(2n-1) + \Lambda(2n+1) - 4C.B(n)] H_s(n).$$

No injectivity assumption on $\mathbf{p} \mapsto P(\mathbf{p})$ is required. This also exposes a source-interface correction: the old unweighted integer P_{sm} has coefficient 1, whereas the physical pushforward carries $\Omega_C^\#(n)$. Therefore $S_{\text{vec}} = P_{\text{sm}}$ is not automatic. The safe current object is the weighted source P_{sm}^Ω unless $\Omega_C^\#(n) = 1$ is separately proved.

1.6 Shared gcd and proof-local Ω : corrected scope

There is no shared-gcd coordinate on the one-copy master index ξ . The legal dependency order is:



Thus any earlier one-copy shared-gcd field is retained only as abstract scaffolding, not as the literal physical dictionary. The abstract dyadic partition remains valid.

1.7 Perron correction and remaining census

For the physical Ford contour with

$$c = 1/\log X, \quad T = X^\delta,$$

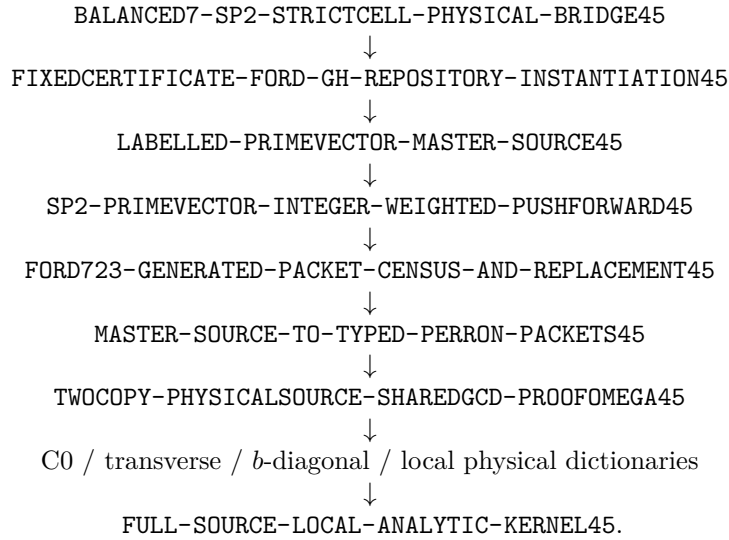
the raw single-contour absolute mass is

$$\int_{-T}^T \frac{d\tau}{\sqrt{c^2 + \tau^2}} = 2 \operatorname{arsinh}(T/c) = O_\delta(\log X),$$

not $O(\log \log X)$. An $O(\log \log X)$ cost corresponds instead to a polylogarithmic vertical height. The current open problem is the exact physical contour-label census and sharing pattern; Ford–Maynard gives a fixed, X -independent paper-level architecture, but the repository-level generated packet family has not yet been exhaustively transcribed and assigned to physical owners.

1.8 Current controlling dependency chain

The source-reconstruction DAG is now:



The first immediate physical bridge binds the strict balanced-seven source data to the repository packet. The next large source/compiler task is the literal Ford-(7.23) generated-packet census and replacement by the already banked source-specific local owners. No local C0/transverse/ b -diagonal theorem is promoted merely from having a compatible shape.

1.9 Status after the 1 September update

Node	Status	Interpretation
Finite $M \leq 4 \times 10^9$ exclusion	KERNEL-PROVED	Unchanged unconditional finite theorem.
Supplied master-source-frontier Aristotle bank	KERNEL-CHECKED BUNDLE	8330 jobs / 0 errors; physical interfaces remain explicit.
Ford G/H source	SOURCE FOUND	Authoritative paper source recovered.
Balanced-seven strict cell / $H_*(P) = -20$	PAPER/RESEARCH PASS	Physical Lean binding still pending.
Labelled prime-vector master source	PAPER/RESEARCH PASS	Correct earliest source index recovered.
Weighted integer pushforward	PAPER/RESEARCH PASS	Carries $\Omega_C^\#(n)$; unweighted match not automatic.
BALANCED7-SP2-STRICTCELL-PHYSICAL-BRIDGE45	FIRST PHYSICAL BRIDGE / OPEN	Bind literal source fields to the repository packet.
FORD723-GENERATED-PACKET-CENSUS-AND-REPLACEMENT45	NEXT SOURCE/COMPILER FRONTIER / OPEN	Exhaust Ford-(7.23) branches and map each packet to a legal owner.
MASTER-SOURCE-TO-TYPED-PERRON-PACKETS45	OPEN	Exact generated packet/contour census not yet complete.
FULL-SOURCE-LOCAL-ANALYTIC-KERNEL45	OPEN	No 100% physical source coverage claim yet.
WindowPairSupply / final large- M bridge	OPEN	End-to-end public-predicate closure not supplied.
Erdős Problem #287	OPEN	No unconditional solution claim.

Publication firewall. The present update materially improves source provenance and proof engineering, but it is not an end-to-end proof. In particular, a source dictionary is not a physical realisation, a fixed-packet theorem is not a full-source theorem, and a conditional compiler does not supply its own analytic or source inputs.

2 31 August 2026 C0/transverse synchronization (historical checkpoint)

This section records the 31-August analytic checkpoint. It is retained as historical provenance; the controlling R9 source/compiler status is Section 1. Earlier R7 uniform- $k = 0$ material remains later in the document as a dated research branch.

2.1 C0 exact product and Double Type II

After canonical reduced-projective conditioning the Kloosterman kernel becomes independent of the balanced b source. For $b = \ell_b d_b e_b$ with $d_b, e_b > Y$, combine $n = \ell_b d_b$. Residue aggregation modulo x and inversion on unit residues give a restricted discrete Fourier matrix of exact operator norm $\sqrt{x}$. With fixed-depth physical product energy,

$$\eta_b^2 \ll L^C \left(\frac{x}{B} + \frac{1}{N} + \frac{1}{E_b} + \frac{1}{x} \right).$$

The product-energy theorem is asserted only for the literal bounded dyadic physical packet; the complete unit branch is $\gcd(ba_\rho b_\rho u_\rho, x) = 1$; nonunit cells belong to D4; and small- x cells route to the short-conductor/low- Q owner. Constants are chosen in the order structure, packet losses, K_* , K_{tr} , (K_Y, K_x) , then k . Thus exact product collision and Double Type II are analytically closed, and C0 is analytically closed conditional on formal normalisation.

2.2 C0 corrections

The old post-conditioning index $n = j\bar{u}' \pmod{x}$ and old wrap $1 + UU'/B$ are **RETRACTED** as controlling C0 architecture. Historical occurrences are preserved. The full physical source is no longer advertised as requiring a moving-level inverse-convolution large sieve after canonical projective cancellation.

2.3 Transverse repaired reciprocity and finite Fourier

For a packet, $R_P = z^2 q_C q_m$ with $q_C = 2(e/a_1)(r_2/c_2)b_1^\flat$, and symmetrically for the primed packet. Full CRT fusion of q_C and q_m leaves cross-modulus inverse coefficients in the numerator, so the naive two-signless DFT is **RETRACTED**.

For

$$\Phi_P = e_q(A\bar{r}\bar{m})e_r(B\bar{q}\bar{m}),$$

reciprocate only the q component. With

$$\Gamma_P = -A + mB\bar{m} \pmod{r}, \quad g_P = (B - A, r), \quad m_P = rm/g_P,$$

one obtains

$$\Phi_P = \text{Arch}_P e_{m_P}(\Gamma_P^{\text{red}} \bar{q}), \quad (\Gamma_P^{\text{red}}, m_P) = 1.$$

This is **THREEFACTOR-TRANSVERSE-ONECONDUCTOR-RECIPROCITY45**.

If $q = S_1 S_2$, the finite operator gives

$$\left| \sum \alpha(S_1) \beta(S_2) e_{m_P}(\Gamma_P^{\text{red}} \overline{S_1 S_2}) \right| \ll L^C \sqrt{m_P(1 + L_1/m_P)(1 + L_2/m_P)} \|\alpha\|_2 \|\beta\|_2,$$

and hence

$$\eta_P^2 \ll L^C \left(\frac{m_P}{L_1 L_2} + \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{m_P} \right).$$

This is **THREEFACTOR-TRANSVERSE-qC-UNITARYFOURIER45**. It accepts arbitrary L^2 vectors and spends no Möbius cancellation. The dense- q_C compiler closes only packets satisfying its explicit dominance/two-long-source-group hypotheses.

2.4 Exact current frontier

The older `THREEFACTOR-TRANSVERSE-CARRIERFACTORIZATION-PAIR-EXISTENCE45` is SUPERSEDED AS FIRST FRONTIER. The first current analytic residual is

`THREEFACTOR-TRANSVERSE-CRITICAL-BEZOUTNUMERATOR-SINGLECARRIER45`.

The surviving operator retains both packet orientations and the full Bézout-dependent reciprocal numerator. It includes nondominant q_C/q'_C cells, partitions with one short side, single atomic-carrier dominance, and critical $q \asymp rm$ geometry. Candidate second Fourier axes include the Bézout numerator, an opposite-packet carrier, the m variable after TT^* , a signed L^2 carrier, grouped short factors, or both orientations. None is promoted here.

The parallel branch

`BDIAGONAL-AFFINEA-SURVIVING-VERTEX-RECT45 : OPEN`

is kept separate.

2.5 Formal/Aristotle firewall

The supplied 31-August bundle records the completed C0 unitary-Fourier / transverse-carrier-interface bank and a default `lake build` of 8298 jobs with zero errors. New declaration axiom audits report only ordinary Mathlib foundations such as `propext`, `Classical.choice`, and `Quot.sound`, with no reported `sorry`, custom axiom, `unsafe`, opaque shortcut, `implemented_by`, or `native_decide`. The named newest files depend on a larger formal graph than the current public-main tree, so the paper records the verified supplied bank without claiming that public `RequestProject/Main.lean` already reproduces the 8298-job build.

3 The problem and the scope of this note

The public problem asks whether, for distinct integers

$$1 < n_1 < \cdots < n_k, \quad \sum_{i=1}^k \frac{1}{n_i} = 1,$$

one necessarily has

$$\max_{1 \leq i < k} (n_{i+1} - n_i) \geq 3.$$

It is listed as open in the Erdős Problems archive and originates in the sources cited there [1, 2].

For a finite set $A \subset \mathbb{N}$, write

$$N = \min A, \quad M = \max A.$$

A counterexample is a finite set satisfying

$$A \subset \{2, 3, \dots\}, \quad \sum_{a \in A} \frac{1}{a} = 1,$$

and the local gap condition that every nonmaximal $a \in A$ is followed by $a + 1$ or $a + 2$ in A . The Lean development formalizes exactly this predicate and proves the ordered form from it.

This note is deliberately not a chronological research diary. It records only the current controlling finite theorem, the exact conditional closure interface, the source-minimal algebra, the factorial Euler balanced-seven reduction, and the current open analytic obligations. Earlier squarefree/collision routers, Vaughan decompositions, failed provider dictionaries, hostile audits, and superseded residuals belong to the separate technical verification ledger and immutable V15 archive.

4 The machine-checked finite theorem

Theorem 4.1 (Finite exclusion). *There is no exact counterexample to Erdős Problem #287 with*

$$3 \leq M \leq 4,000,000,000.$$

Equivalently, every unit-fraction representation in the problem with maximum denominator at most 4×10^9 has a consecutive gap at least 3.

The theorem is represented in the repository by `no_Erdos287Counterexample_of_max_le_4e9`. Its proof is not an exhaustive enumeration of finite subsets. It is a certificate compiler built from the three ideas below.

4.1 Holes and the adjacent-hole contradiction

For a hypothetical counterexample, call an integer $x \in [N, M]$ a *hole* when $x \notin A$. The $\text{gap} \leq 2$ condition implies that two consecutive integers in $[N, M]$ cannot both be holes. This is the formal theorem `Gap2CE.holes_isolated`.

A second elementary fact places the top half of the interval above the minimum:

$$N \leq \left\lfloor \frac{M}{2} \right\rfloor.$$

Thus, whenever $M \leq 2x$ and $x + 1 \leq M$, the consecutive pair $x, x + 1$ lies inside the closed window where the adjacent-hole contradiction applies.

4.2 Prime-power windows

For $j \geq 0$, let $C(j)$ denote the largest reduced numerator arising from a nonempty partial reciprocal sum

$$\sum_{s \in S} \frac{1}{s}, \quad \emptyset \neq S \subseteq \{1, \dots, j\}.$$

The formal development uses a certified upper table

$$C_{\text{val}}(0), \dots, C_{\text{val}}(9) = 1, 1, 3, 11, 25, 137, 137, 1019, 2143, 7129.$$

The prime-power window theorem says, schematically, that if p is prime, $e \geq 1$, $p^e \mid x$, and

$$C\left(\left\lfloor \frac{M}{p^e} \right\rfloor\right) < p,$$

then x cannot belong to A . The reason is that the top p -adic layer of the reciprocal sum would force a nonzero numerator to vanish modulo p .

Consequently, if both x and $x + 1$ possess such prime-power divisors, then they are two adjacent holes. This gives the central finite compiler.

Proposition 4.2 (Window-pair blocker). *Suppose there are primes p_u, p_v , positive exponents a_u, a_v , and an integer x such that*

$$\begin{aligned} p_u^{a_u} \mid x, \quad p_v^{a_v} \mid x + 1, \\ \left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor, \left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor \leq 9, \\ C_{\text{val}}\left(\left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor\right) < p_u, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor\right) < p_v, \end{aligned}$$

and

$$M \leq 2x, \quad x + 1 \leq M.$$

Then a $\text{gap} \leq 2$ counterexample with maximum M cannot exist.

4.3 The interval-certificate chain

One certificate can exclude an entire interval $L \leq M \leq U$. The divisibility data are fixed at a chosen x , monotonicity controls the windows as M varies, and $U \leq 2x$ keeps the forced holes above the minimum. The repository contains a chain of 34 certificates whose intervals meet end-to-end and cover

$$[3, 4 \times 10^9].$$

All primality side conditions are discharged in the kernel; the construction does not use `native_decide`.

4.4 Sign-sensitive Sophie witnesses and the free maximum blocker

The latest supplied finite bank sharpens the final analytic-to-finite landing and is worth stating because it prevents small endpoint mistakes later. A *plus* Sophie witness at M is a prime q with

$$q \geq 5, \quad M < 3q, \quad 2q + 1 \leq M, \quad 2q + 1 \text{ prime},$$

equivalently $M/3 < q \leq (M-1)/2$ with the integer endpoints understood literally. A *minus* witness is

$$q \geq 5, \quad M < 3q, \quad 2q \leq M, \quad 2q - 1 \text{ prime},$$

so the correct minus endpoint is

$$q \leq \left\lfloor \frac{M}{2} \right\rfloor.$$

Either witness contradicts an exact counterexample. The hypothesis $q \geq 5$ is not cosmetic: for a prime it is the same as $q > 3$, which is the finite window threshold needed by the cofactor-two obstruction.

There is also a free maximum-divisor blocker. If $q > 3$ is prime, $q \mid M$ and $M < 3q$, then necessarily $M \in \{q, 2q\}$; in either case the top-layer/cofactor-two exclusion contradicts $M \in A$. Thus a counterexample cannot have prime maximum and cannot have $M = 2q$ with $q > 3$ prime. These are finite repairs, not analytic inputs; they should be used when translating positive affine mass into an actual blocker.

5 The exact large- M closure criterion

Definition 5.1 (Window-pair supply). For $M \in \mathbb{N}$, $\text{WindowPairSupply}(M)$ is the assertion that there exist $x, p_u, a_u, p_v, a_v \in \mathbb{N}$ such that

$$\begin{aligned} p_u, p_v \text{ are prime,} \quad a_u, a_v \geq 1, \\ p_u^{a_u} \mid x, \quad p_v^{a_v} \mid x + 1, \\ \left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor \leq 9, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor\right) < p_u, \\ \left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor \leq 9, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor\right) < p_v, \\ M \leq 2x, \quad x + 1 \leq M. \end{aligned}$$

In plain terms, two consecutive integers in the top half of $[1, M]$ each have a prime-power divisor larger than roughly $M/10$, with a small local numerator condition.

Theorem 5.2 (End-to-end conditional compiler). *Assume there is an explicit $M_0 \leq 4 \times 10^9$ such that*

$$\text{WindowPairSupply}(M) \quad \text{for every } M \geq M_0.$$

Then Erdős Problem #287 is true.

This implication is machine-checked as `no_Erdos287Counterexample_of_closure`. The proof splits at 4×10^9 : the finite theorem handles the lower range, while the window-pair blocker handles every larger maximum.

The interface is weaker than a Sophie-Germain prime condition. For example, if a prime q in the interval $M/3 < q \leq (M - 1)/2$ has $2q + 1$ prime, then one may take $x = 2q$. The analogous minus case uses $x = 2q - 1$. The Lean theorem `windowPairSupply_of_sophieWitness` formalizes this implication. The present programme does not claim such Sophie witnesses for every large M .

Part I

Theory and Verification Manual: Gate 0–2

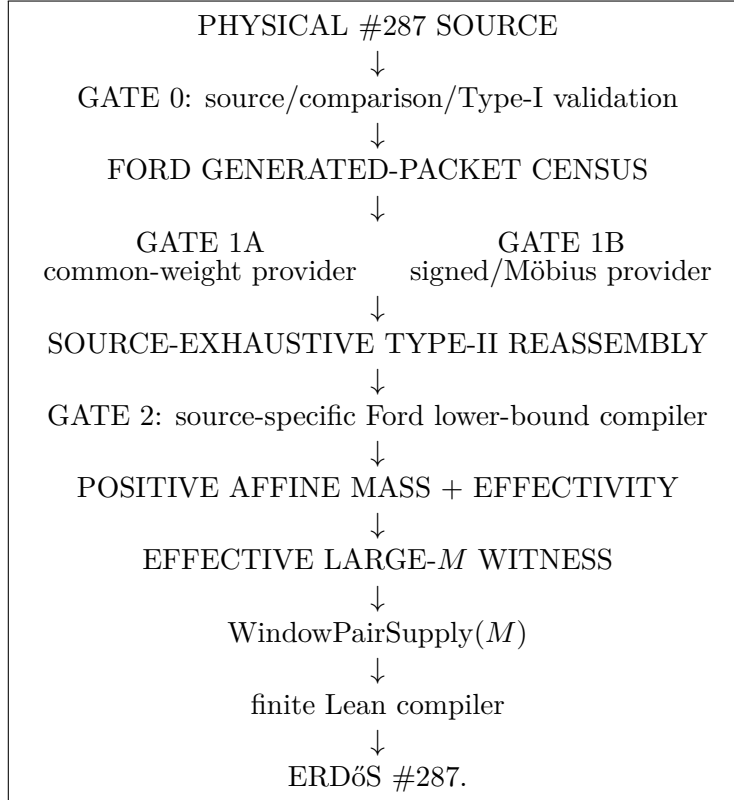
6 What the Gate theory is trying to do

This Part is written for a reader who has never seen the earlier chats, the V15 patch stack, or the repository. The Gate labels are not theorem substitutes: each Gate is a literal mathematical package with an input source, a target theorem, a verification boundary, and specified downstream consequences.

Conceptually, the programme separates four jobs.

- Gate 0.** Construct and validate the physical sequence/comparison class: prime mass, comparison regularity, growth, Type I, and the exact expected-term convention.
- Gate 1.** Supply parity-breaking Type-II estimates for the actual generated packets.
- Gate 1A.** One canonical common-weight moving-source provider.
- Gate 1B.** A complementary signed/Möbius/high-order provider that preserves the arithmetic sign resource through completion/dispersion.
- Gate 2.** Consume a source-exhaustive Type-II package, the Ford arithmetic weight, the exceptional N_2 sector, and one compatible ε_* to run the Ford–Maynard lower-bound endgame.

The intended dependency graph is



(Theory-DAG)

This is a dependency graph, not a proof.

The following non-implications are load-bearing:

Gate 1A closed $\nRightarrow$ Gate 1B closed, Gate 1B closed $\nRightarrow$ Gate 1A closed, Gate 1A + Gate 1B $\nRightarrow$ Full Type II without source exhaustiveness, generated Type II $\nRightarrow$ Gate 2 without Gate-0/ N_2 /source inputs, Balanced7 closed $\nRightarrow$ FCL closed, FCL closed $\nRightarrow$ WindowPairSupply(M), none of the preceding statements alone proves Erdős #287.	(Theory-firewall)
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6.1 Verification vocabulary

Every major object below is assigned one of the following meanings.

Label	Meaning
LEAN-PROVED	Proved by a cited/current Lean kernel replay.
PROVED-FINITE / PROVED-ALGEBRAIC	Exact finite/combinatorial/algebraic mathematics; may or may not also be Lean formalized.
EXTERNALLY-AUDITED	A sourced mathematical theorem or source audit, not formalized here.
CONDITIONAL-COMPILER	The implication is proved, but one or more explicit antecedents remain open.
SOURCE-OPEN	The physical/source identification, normalization or dictionary is missing.
ANALYTIC-OPEN	The source object is sufficiently fixed; a deep estimate remains unproved.
SOURCE-BLOCKED	The literal source object/prefactor needed even to state the final theorem is not yet printed.
CAPACITY-ONLY	Exponent arithmetic passes, but no analytic theorem follows.
FALSE / RETIRED	The literal proposed route is disproved or withdrawn.
SUPERSEDED	Historically valid but no longer the controlling representation.

7 Gate 0: the physical sequence/comparison theorem

7.1 The physical #287 sequence

Fix a nonnegative smooth weight W supported inside the analytic blocker band; the finite blocker itself only needs the corresponding interval to lie in the top half. Define

$$\mathfrak{S}_2 = \prod_{\ell > 2} \frac{\ell(\ell-2)}{(\ell-1)^2}, \quad B(n) = \mathfrak{S}_2 \prod_{\substack{\ell|n \\ \ell > 2}} \frac{\ell-1}{\ell-2},$$

and

$$a_X(n) = W(n/X)(\Lambda(2n-1) + \Lambda(2n+1)),$$

$$b_X(n) = 4W(n/X)B(n), \quad w_X(n) = a_X(n) - b_X(n). \quad (\text{G0-source})$$

The role of b_X is to model the expected local density of the two affine prime forms; w_X is the discrepancy on which Type I/II cancellation is sought.

The elementary comparison algebra includes

$$B(n) = \mathfrak{S}_2 \sum_{\substack{e|n \\ e \text{ odd squarefree}}} \prod_{\ell|e} \frac{1}{\ell-2},$$

and for each fixed d ,

$$\lim_{Y \rightarrow \infty} \frac{1}{Y} \sum_{m \leq Y} B(dm) = \frac{d}{\phi(2d)}. \quad (\text{G0-Bmean})$$

Thus one sign has mean $2d/\phi(2d)$ and the two-sign model has mean $4d/\phi(2d)$ at the fixed- d level. Also

$$B(n) \ll \log \log(3n), \quad |w_X(n)| \ll \log X,$$

and, in the present source class,

$$\sum_{n \lesssim X} |w_X(n)| \tau(n) \ll X (\log X)^2. \quad (\text{G0-growth})$$

7.2 Gate-0 target package

Definition 7.1 (Gate-0 package). Gate 0 is *closed for the #287 source at scale X* if the following seven items hold with one common physical normalization.

G0.1 $a_X(n) \geq 0$ and the comparison sequence b_X belongs to the Ford–Maynard ordinary sequence class used in the lower-bound theorem.

G0.2 Prime mass:

$$\sum_p b_X(p) \gg \frac{X}{\log X}.$$

G0.3 Growth/divisor control, including a bound of the form (G0-growth) (or the exact stronger divisor-weight norm required by the chosen Ford invocation).

G0.4 Comparison regularity: the physical b_X satisfies the Ford–Maynard condition (b.2) uniformly over every coagulation region actually used downstream. At the fixed- d level this is compatible with (G0-Bmean); the required uniform publication-level statement must be source-pinned.

G0.5 Literal maximal/weighted Type I:

$$\sum_{m \leq X^{1/2-\varepsilon}} \tau(m)^A \max_I \left| \sum_{\substack{X/2 < mn \leq X \\ n \in I}} w_X(mn) \right| \ll_{A,W,\varepsilon} X (\log X)^{-A}. \quad (\text{G0-I})$$

G0.6 One fixed principal/small-conductor/exceptional-character expected-term convention is used consistently by both the prime source and the comparison source.

G0.7 Even, nonunit, prime-power and boundary strata created by the affine forms $2n \pm 1$ are either included in the source theorem or routed with a proved error.

Current verification. Items G0.1–G0.3 have elementary/source-level proofs in the dossier; the exact finite problem statement and finite blocker are Lean-proved. G0.4 and G0.5 are **SOURCE-OPEN / EXTERNALLY-AUDITED TARGETS** at the exact publication normalization; G0.6 is the same physical comparison convention that later appears in the factorial balanced-seven and Pure5 comparison pins.

What Gate 0 closing buys. It supplies the non-parity hypotheses needed by the Ford lower-bound compiler and removes the possibility that a later Type-II estimate is being applied to the wrong comparison sequence.

What it does not buy. Gate 0 does not prove Type II, parity breaking, Gate 1A, Gate 1B, Gate 2, a large- M witness, or Erdős #287.

8 Gate 1A: canonical common-weight Type-II provider

8.1 Parameters, rows and physical source

At an auxiliary scale X_1 , set

$$\begin{aligned} M &= X_1^{1/3}, & R &= X_1^a, & L &= X_1^b, \\ H &= X_1^{a+2b-2/3}, & K &= \frac{M}{R}, & U &= \frac{L}{H}, & D &= \frac{L^2}{H}. \end{aligned} \quad (\text{G1A-scales})$$

Thus

$$RK = M, \quad DH = L^2, \quad M^2H = RL^2.$$

A row is

$$e = (r, k, m), \quad r \asymp R \text{ prime}, \quad 1 \leq |k| \leq K, \quad m \asymp M, \quad m' = m + kr \asymp M.$$

Choose the canonical affine forms $A_e(t), B_e(t)$ so that

$$\boxed{m'A_e(t) - mB_e(t) = 2k.} \quad (\text{G1A-det})$$

For

$$\rho_\ell(y) = \mathbf{1}_{\ell|y} - \frac{1}{\ell}$$

and bounded prime coefficients b_p, d_q , the *one common physical-weight* source is

$$\boxed{C_e(b, d) = \sum_{t \in \mathbb{Z}} W_D(t) \left(\sum_{p \asymp L} b_p \rho_p(A_e(t)) \right) \left(\sum_{q \asymp L} d_q \rho_q(B_e(t)) \right).} \quad (\text{G1A-source})$$

The word “common” is literal: W_D is one datum. Arbitrary independently chosen weights $W_{D,e}$ are a stronger source class and are not covered by the canonical theorem.

8.2 Gate-1A target theorem

Theorem 8.1 (Canonical common-weight Gate-1A provider – conditional form). *Assume the five source/analytic pins G1A-A through G1A-E below. Then, uniformly for the canonical all- m row family $\mathcal{E}^\sharp$,*

$$\boxed{\sum_{e \in \mathcal{E}^\sharp} |C_e(b, d)|^2 \ll \frac{ML^4}{H} X_1^{o(1)} \left(\sum_{p \asymp L} |b_p|^2 \right) \left(\sum_{q \asymp L} |d_q|^2 \right).} \quad (\text{G1A-target})$$

The exact normalized variant used by the formal compiler is equivalent after the banked H^2 normalization bridge.

The theorem is a **CONDITIONAL-COMPILER**: the finite/operator part is substantially banked, but the physical source pins are not all inhabited.

8.3 The five load-bearing pins

G1A-A: Poisson–Bruhat/source transport. After the one-sided physical transform and reciprocity, every retained source state ξ must have moving-prime coefficient

$$c_r(\xi) = B_\xi F_\xi(r/R), \quad (\text{G1A-PB})$$

where F_ξ is a uniformly smooth envelope generated from the one physical W_D , not an arbitrary row weight. The transport must preserve the physical/normalized energy up to a square-summable source error.

STATUS: SOURCE-OPEN / CONDITIONAL PIN.

Without this pin, the later moving-prime participation argument would apply to a convenient model rather than to the physical packet.

G1A-B: absolute source ceiling. The transported physical source must satisfy

$$\mathcal{T}_{\text{abs}} \ll M^2 L^4 X_1^{o(1)}. \quad (\text{G1A-abs})$$

This is the absolute comparator against which the moving-family contraction is measured.

STATUS: SOURCE/ANALYTIC PIN.

G1A-C: fixed-state exclusion and prime participation. A fixed nonprojective state must not absorb the moving prime variable. Smoothness of F_ξ produces a long plateau; the short-interval prime theorem then gives

$$N_\xi \gg R^{3/4-o(1)} \quad (\text{G1A-part})$$

participating prime rows for each retained state. The finite BPP compiler converts this participation into

$$E_{\text{np}} \ll R^{-1/2+o(1)} \mathcal{T}_{\text{abs}}. \quad (\text{G1A-np})$$

The project kernel-checks the finite participation-to-energy implication; it does not formalize the entire physical smooth-envelope/short-interval input.

Status. FINITE COMPILER LEAN-PROVED; the physical participation statement is EXTERNAL / SOURCE-OPEN.

G1A-D: outer-root physical energy and projective routing. The abstract nonprojective energy must be identified with the physical Gate energy, and the projective/axis/diagonal branch must be routed in the same source normalization. The finite ratio-class algebra includes the state diagonal in the zero-curvature branch, and the physical projective mass target is

$$E_{\text{proj}} \ll MHL^4 X_1^{o(1)}. \quad (\text{G1A-proj})$$

Exactly one outer square root converts (G1A-np) to an $R^{-1/4+o(1)}$ contraction. At the three controlling vertices the signed margins are

$$\frac{1}{72}, \quad \frac{1}{24}, \quad \frac{1}{32}. \quad (\text{G1A-margins})$$

Status. FINITE SQUARE-ROOT / PROJECTIVE ALGEBRA BANKED; the physical energy identification is SOURCE-OPEN.

G1A-E: recombination/discarded-sector bound. After all source truncations, exceptional routes and recombinations, the physical error must satisfy

$$E_{\text{rec}} \ll U^{-2+o(1)} M^2 L^4, \quad (\text{G1A-rec})$$

with the source-exact prefactor and without spending the same saving twice. The exponent margins at the controlling vertices are

$$\frac{1}{18}, \quad \frac{1}{18}, \quad \frac{1}{24}.$$

Status. EXPONENT BUDGET BANKED; literal source recombination is SOURCE-OPEN.

8.4 What the BPP mechanism proves and what it does not

The finite BPP mechanism is a local-to-family compiler:

$$\begin{array}{c} \text{smooth moving-prime participation} \\ \Downarrow \\ R^{-1/2+o(1)} \text{ family-energy contraction} \\ \Downarrow \\ R^{-1/4+o(1)} \text{ after one outer root.} \end{array}$$

It does *not* prove the source transport, absolute source ceiling, physical energy identification or recombination estimate. Those are separate statements precisely so that a reader can attack or falsify them independently.

8.5 The source-glue/SB- ν problem

The current source-glue question is whether the same physical packet can simultaneously satisfy the common-weight representation, the BPP state representation and the final routing representation with fibre multiplicity

$$N_\nu = X_1^{o(1)} \quad (\text{G1A-SBnu})$$

in the synchronized two-view model. This is a source-identification frontier rather than a new exponent computation.

Finally,

$$\text{GATE1A_REQUIRED_FOR_287} = \text{UNKNOWN.}$$

The Ford-generated packet census, not programme nomenclature, decides whether a needed #287 packet truly belongs to the canonical common-weight source class.

9 Gate 1B: signed/Möbius parity-breaking provider

9.1 Native centering, sign and complementary-divisor geometry

For coprime d, p ,

$$\rho_{dp}(y) = \rho_d(y)\rho_p(y) + \frac{\rho_d(y)}{p} + \frac{\rho_p(y)}{d}. \quad (\text{G1B-center})$$

If q is squarefree and $p \mid q$ is the distinguished prime, then

$$\mu(q/p) = -\mu(q). \quad (\text{G1B-sign})$$

This is the sign resource that must not be squared away accidentally.

The native complementary-divisor source exposes

$$\boxed{q\ell - uv = 2.} \quad (\text{G1B-det})$$

This determinant-two geometry is separate from the Gate-1A determinant $2k$ and from the #287 $\mu * \log$ determinant-one line.

9.2 The banked one-completion source

On dyadic windows $d \sim D$ and $p \sim P$, the authoritative one-completion coefficient is

$$\boxed{\beta_{D,P}(d, p) = \mu_D(d) \Lambda_P(p),} \quad (\text{G1B-beta})$$

where μ_D, Λ_P are the physical truncated Möbius and prime weights on the stated windows. The public formal challenge proves the separate linearity/bilinearity of this coefficient and an exact level-of-distribution budget requiring

$$\boxed{\theta \geq \frac{1}{2} + \frac{1}{104}.} \quad (\text{G1B-104})$$

It also proves the implication from an inhabited source-to-convolution-BV interface to the abstract Gate conclusion. Producing that interface from the literal physical source remains open. This is a valid formal provider socket, but it is no longer the first research-level analytic obstruction.

9.3 The Motohashi line

Fix u, ℓ and a base solution to

$$\boxed{uv_0 + 2 = \ell z_0.} \quad (\text{G1B-base})$$

All solutions on the complementary-divisor line are

$$\boxed{v_t = v_0 + \ell t, \quad z_t = z_0 + ut.} \quad (\text{G1B-line})$$

Let H be the physical line length and write $e_H(x) = e^{2\pi i x/H}$. For the exact one-completion coefficient and a smooth cutoff W_1 define

$$\boxed{\mathcal{Z}_{u,\ell}(k) = \sum_t \beta_{D,P}(z_t) W_1(t/H) e_H(-kt).} \quad (\text{G1B-Z})$$

For the five centred defect factors

$$b_5 = \delta_1 * \delta_2 * \delta_3 * \delta_4 * \delta_5$$

and a second physical cutoff W_2 , define

$$\boxed{\mathcal{B}_{u,\ell}(k) = \sum_t b_5(v_t) W_2(t/H) e_H(-kt).} \quad (\text{G1B-B})$$

The Motohashi pairing is

$$\boxed{\mathcal{M}_{u,\ell} = \frac{1}{H} \sum_k \mathcal{Z}_{u,\ell}(k) \overline{\mathcal{B}_{u,\ell}(k)}.} \quad (\text{G1B-M})$$

The current supplied research ledger reports the exact A/B/C source pin, the twisted-defect A/B/C reduction and the fivefold Motohashi iteration as passed source algebra; the fixed-polylogarithmic- k interior is closed modulo the comparison convention.

9.4 Endpoint residue coefficient and diagonal/off-diagonal split

At the endpoint define, for a unit residue $a \bmod \ell$,

$$A_{\ell,a,k} = \sum_{u \equiv -2a^{-1} \pmod{\ell}} a_4(u) \mathcal{Z}_{u,\ell}(k), \quad (\text{G1B-A})$$

where a_4 is the four-defect coefficient inherited from the physical source. The endpoint Cauchy energy is

$$\mathcal{E}_{\text{end}}(k) = \sum_{\ell \sim L} \sum_{a \in (\mathbb{Z}/\ell\mathbb{Z})^\times} |A_{\ell,a,k}|^2. \quad (\text{G1B-Eend})$$

Opening the square gives

$$\mathcal{E}_{\text{end}}(k) = \sum_{\ell,a} \sum_{u_1, u_2 \equiv -2a^{-1} \pmod{\ell}} a_4(u_1) \overline{a_4(u_2)} \mathcal{Z}_{u_1,\ell}(k) \overline{\mathcal{Z}_{u_2,\ell}(k)}.$$

The diagonal is $u_1 = u_2$. The off-diagonal is parametrized by

$$u_2 = u_1 + j\ell, \quad j \neq 0. \quad (\text{G1B-j})$$

For corresponding line parameters t_1, t_2 ,

$$z_2(t_2) - z_1(t_1) = u(t_2 - t_1) + j v_{t_2}, \quad (\text{G1B-off-source})$$

and, after opening $\beta = \mu_D \Lambda_P$,

$$d_2 p_2 - d_1 p_1 = u(t_2 - t_1) + j v_{t_2}. \quad (\text{G1B-off-beta})$$

These identities are the literal arithmetic object behind the current endpoint off-diagonal problem.

The latest supplied research ledger records

$$\text{endpoint } u\text{-diagonal} = \text{power closed with } X^{-1/12+o(1)}. \quad (\text{G1B-diag-status})$$

9.5 Current Gate-1B research frontier: Bézout dual core

The previous repair label **SHIFTED-MAM-TOPBAND45** and the earlier endpoint label **RANKONE-ENDPOINT-U-OFFDIAG45** remain useful provenance, but neither is the current first exact Gate-1B analytic residual. The later centred-AP/Bézout reduction strictly reduces the parent **C4SHIFT-OFFDIAG-CENTERED-AP5/8-GRAM45** to

$$\boxed{\text{C4SHIFT-BEZOUT-DUAL-ROELL45}}.$$

Its status is **ANALYTIC OPEN / FIRST EXACT GATE-1B RESIDUAL AT RESEARCH LEVEL**; the quantitative D -census promotion feeding this node remains **HOSTILE-AUDIT PENDING**.

On the clean sector

$$(d, \ell) = 1, \quad r = dr_0,$$

the physical Bézout line is

$$\boxed{bZ_2 - aZ_1 = \ell r_0},$$

with parametrization

$$Z_1 = \ell r_0 \kappa_{a,b} + bt, \quad Z_2 = \ell r_0 \bar{b}_a + at.$$

The physical AP residue is

$$\boxed{t \equiv -2\overline{usdab} \pmod{\ell}},$$

and is independent of r_0 . After cyclic dualization the centred kernel is

$$\widehat{K}_{\text{cent}}(\nu) = \frac{e_\ell(-\nu\tau)}{\ell} \mathfrak{q}_{\ell, \xi_1, \xi_2, dab}(\nu),$$

where

$$\mathfrak{q}(\nu) = 1 - \frac{1}{\ell} \sum_{k \bmod \ell} \mathcal{M}_1(k + \nu \overline{dab}) \overline{\mathcal{M}_2(k)}.$$

The zero dual mode is a *local owner*, not zero; it belongs to the parallel source comparison child

$$\boxed{\text{C4SHIFT-BEZOUT-ZEROMODE-LOCALMATCH45: \quad SOURCE \quad OPEN}},$$

itself inside TOPBAND-BROAD-MAJOR-TREE-MATCH45.

For nonzero dual frequency the analytic phase is

$$e_\ell\left(2\nu\overline{usdab}\right).$$

Opening the two-coordinate gamma source gives

$$bx_2y_2 - ax_1y_1 = \ell r_0, \quad e_\ell(\nu t) = e_\ell(\nu \bar{b} x_1 y_1).$$

The latest research ledger proposes the D -local estimate

$$\mathcal{N}_D \ll Y^{3/4} \sqrt{\min(D, T/D)} L^C, \quad T = Y^{3/2} L^{O(1)}.$$

This closes the boundary D -bands provisionally, but the central range remains. At the worst balanced point $D = T^{1/2} = Y^{3/4}$ the provisional bound leaves residual

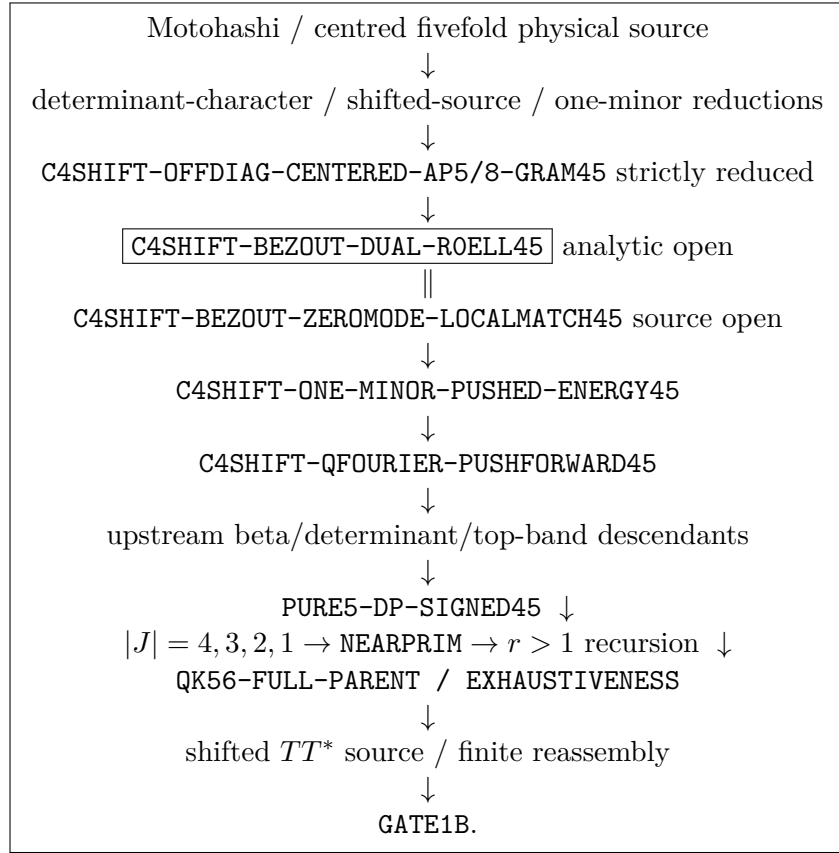
$$Y^{3/8} = X^{1/24}.$$

The fixed- (d, a, b, ℓ) uniqueness/census used to reach this statement must still be hostile-audited with the possibility that two admissible t values differ by a multiple of ℓ . No literal Kloosterman/spectral provider is currently matched.

Thus the missing Gate-1B mechanism is joint cancellation of the nonzero t -dual reciprocal phase against the exact shifted gamma Bézout source while retaining the coupled (ν, r_0, ℓ) family. Gate 1B remains open.

9.6 Downstream Gate-1B theorem objects

The current Gate-1B firewall is



The zero dual mode/local-match branch may not be skipped. The provisional D -local bound is not promoted until its census passes hostile audit, and no downstream node is closed merely by naming the Bézout dual residual.

10 Ford–Maynard generated packets and proof-specific Type II

10.1 Published universal wrapper

The Ford–Maynard lower-bound framework uses a universal Type-II hypothesis. In the present 1/6 specialization a safe sufficient form is

$$\left| \sum_{\substack{(X/2)^\varepsilon < m \leq X^{1/6-\varepsilon} \\ X/2 < mn \leq X}} \xi_m \kappa_n w_X(mn) \right| \ll_A X (\log X)^{-A} \quad (\text{FM-universal-II})$$

for every allowed divisor-bounded coefficient pair. The project does *not* claim that Ford–Maynard publish the weaker proof-specific theorem described next.

10.2 What their proof actually generates

The Ford arithmetic source has

$$G(m; n) = \mathbf{1}_{m \leq n^\gamma} \lambda(m_1) g(\mathbf{v}(m_2; n)), \quad H(n) = \sum_{d|n} G(d; n), \quad (\text{FM-GH})$$

with the canonical rough/smooth split $m = m_1 m_2$. After finite decomposition and condition removal, Proposition 7.22 reaches a fixed-depth product sum (equation (7.23) in the source)

$$\sum_{n=n_1 \cdots n_N \asymp x} w_n x_1(n_1) \cdots x_N(n_N), \quad (\text{FM-7.23})$$

and chooses a product $\prod_{e \in E} n_e$ in the Type-II range. Grouping the selected factors and their complement gives

$$\boxed{\sum_{n=n' n'' \asymp x} Y_1(n') Y_2(n'') w_{n' n''}} \quad (\text{FM-generated-II})$$

The published proof then invokes the universal Type-II hypothesis.

Hence an invocation-by-invocation replacement theorem would be enough for this proof:

every actual generated (FM-generated-II) packet has a proved source-exact provider $\Downarrow$ every Type-II invocation in Proposition 7.22 is covered	(FM-replacement-logic)
--	------------------------

This implication additionally requires a source-exhaustive census and a replacement certificate; it is not obtained merely by identifying some packets.

10.3 The three source obligations

Definition 10.1 (SOURCE-ADAPTER45). For every generated packet, print the literal operator, physical ranges, coefficient class, weight dependence, centering/expected term, gcd/collision restrictions, original prefactor and normalized target, and map it to a proved provider without changing the source class.

Definition 10.2 (FM-TO-GATE-COORDINATE-CENSUS45). Enumerate every Type-II packet actually generated by the Ford decomposition, with enough coordinate information to decide whether it belongs to Gate 1A, Gate 1B or a direct exceptional/standard class.

Definition 10.3 (C_{FM} replacement certificate). Verify invocation by invocation that the generated census covers every use of the universal Type-II hypothesis in the chosen Ford proof, with no omitted packet, no multiplicity loss and no double-spending of an analytic saving.

All three remain open source/reassembly obligations.

10.4 Provider-facing generated families

Family	Literal source/scale	Candidate provider	Why it matches / what is missing
Möbius singleton	one selected $m < X^{1/6}$ fac- tor carries Möbius weight; complement remains generated	Gate 1B signed source	Möbius sign is in the cor- rect physical slot only after a literal completion dictionary; generated outer-factor stabil- ity is still required.
Model singleton	selected factor is model/Mellin-like; Möbius carrier may remain in the complement	QK56 or direct affine completion	provider cannot be inferred from the word “singleton”; na- tive packet dictionary is open.

Family	Literal source/scale	Candidate provider	Why it matches / what is missing
Balanced-seven V-cell	seven fixed prime boxes at $Y = X^{1/7}$	factorial- Ω_7 source, then the fixed-degree-seven attack	factorial Euler encoding is exact; endpoint/3221 analytic theorem and comparison remain.
Pure5/high-order family	five centred defect factors with distinguished-prime/Möbius source	Gate 1B Motohashi/rank-one chain	current first analytic object is the endpoint off-diagonal; high- k and downstream defect/conductor families remain.
Balanced R_9	nine-coordinate Ford-generated leakage cell	fixed-certificate/FCL lane	finite binomial 70 is proved; literal physical $G(d; n)$ specialization is source-open.
Common-weight packets	actual generated source must preserve one physical W_D after scale conversion	Gate 1A	requires exact scale/source dictionary and all five Gate-1A pins.
Remaining fixed- g_* packets	not yet exhaustively censused	TBD from literal source	the finite census/provider assignment itself is still open.

11 Balanced R_9 , fixed-certificate leakage and FCL

The balanced order-nine finite identity is

$$\sum_{j=0}^4 (-1)^j \binom{9}{j} = 70. \quad (\text{R9-70})$$

This is PROVED-FINITE; by itself it is not the physical Ford statement $H(n) = 70$.

To obtain $H(n) = 70$ one must prove that the literal Ford specialization of $G(d; n)$ on the balanced R_9 cell selects exactly the corresponding low/high divisor orders and has the required value of $g(\emptyset)$. That is

$$\boxed{\text{literal } G(d; n) \text{ specialization: SOURCE-OPEN.}} \quad (\text{R9-G-open})$$

Therefore balanced R_9 is a *Ford-generated leakage candidate*, not a completed leakage theorem.

For one fixed positive Ford certificate g_* , define its arithmetic weight

$$H_*(n) = \sum_{d|n} G_*(d; n).$$

Let $\mathcal{P}$ be the desired prime region, $\mathcal{N}_1$ the certificate-sign composite region, $\mathcal{N}_2$ the exceptional collar, and

$$\mathcal{U}_* = (X/2, X] \setminus (\mathcal{P} \cup \mathcal{N}_1 \cup \mathcal{N}_2).$$

The fixed-certificate leakage is

$$\boxed{\mathcal{L}_*(X) = \sum_{n \in \mathcal{U}_*} w_X(n) H_*(n).} \quad (\text{FCL})$$

The N_2 error is

$$\mathcal{E}_{N_2}(X) = \sum_{n \in \mathcal{N}_2} w_X(n) H_*(n).$$

If the total correlation, leakage, N_2 error and comparison error are all small relative to the positive certificate margin $C_*(\varepsilon_*)$, then the fixed-certificate transference gives positive affine prime mass. A sufficient quantitative form is

$$|\mathcal{L}_*(X)| + |\mathcal{E}_{N_2}(X)| \leq \frac{C_*(\varepsilon_*)}{4} B_X, \quad (\text{FCL-margin})$$

together with the corresponding total-correlation/comparison bounds. This is a conditional compiler; it does not itself prove FCL.

Balanced R_9 may contribute to $\mathcal{L}_*$ once the source obligation (R9-G-open) is discharged, but closing that one cell does not close the full leakage family.

12 Gate 2: source-specific Ford–Maynard lower-bound compiler

Fix one compatible

$$P_{\varepsilon_*} = \left(\frac{1}{2} - \varepsilon_*, \varepsilon_*, \frac{1}{6} - 2\varepsilon_* \right).$$

Gate 2 is the theorem-shaped endgame package:

Theorem 12.1 (Gate-2 lower-bound compiler – conditional). *Assume:*

G2.1 *the Gate-0 source/comparison/Type-I package;*

G2.2 *the literal Ford $G(d;n)/H(n)$ source and its sign properties;*

G2.3 *a source-specific aggregate N_2 estimate;*

G2.4 *one common fixed $\varepsilon_* > 0$ compatible with every source range;*

G2.5 *either the universal Type-II theorem (FM-universal-II), or a complete generated-packet provider package together with **SOURCE-ADAPTER45**, **FM-TO-GATE-COORDINATE-CENSUS45** and C_{FM} ;*

G2.6 *the ordinary/bounded-class firewall and every comparison/exceptional-character normalization needed by the chosen Ford theorem;*

G2.7 *effective versions of all inputs at a computable threshold.*

Then the Ford–Maynard lower-bound geometry yields positive affine mass of the form

$$\boxed{\sum_{p \in \mathcal{P}} a_X(p) \geq (C_*(\varepsilon_*) + o(1)) B_X \gg \frac{X}{\log X}}, \quad (\text{G2-positive})$$

for the physical #287 source and the chosen certificate.

STATUS: CONDITIONAL-COMPILER / NOT ACTIVATED.

The theorem still does not by itself produce **WindowPairSupply**. One additional effective witness extraction theorem must convert the positive affine mass into a Sophie/safe-prime or prime-power blocker configuration, or directly into the **WindowPairSupply**(M) predicate of the finite compiler. That final conversion/effectivity step remains open.

13 Current #287 specialization: Balanced7 bank and the uniform $k = 0$ frontier

13.1 Balanced7: current post-audit bank

The controlling fixed-certificate source is the SP-2 smooth-parity packet

$$G_*(m; n) = \mathbf{1}_{m \leq n^{\gamma*}} \mu(m_1) g_*(\mathbf{v}(m_2; n)), \quad H_*(n) = \sum_{m|n} G_*(m; n).$$

On the balanced seven-prime cell

$$P = p_1 \cdots p_7, \quad p_i \asymp X^{1/7},$$

the source coefficient is

$$H_*(P) = 1 - 7 + 21 - 35 = -20.$$

The literal one-sign packet is

$$-20 \sum_{\mathbf{p}} \Omega_{\lambda}(\mathbf{p}) [\Lambda(2P + s) - 2B(P)], \quad s = \pm 1.$$

The source-minimal factorization

$$\Lambda(2P + s) = \sum_{qr=2P+s} \mu(q) \log r$$

forces the physical logarithmic source ledger

$$C_{\text{ext}} = 1;$$

the earlier $C_{\text{ext}} = 0$ claim is superseded.

Full- q Euler principal owner. Define

$$J_P(z) = \sum_{\substack{q \leq z \\ (q, 2P)=1}} \frac{\mu(q)}{\phi(q)} \log \frac{z}{q}.$$

The audited Euler factorization is

$$F_P(w) = \frac{H_P(w)}{\zeta(1+w)}, \quad H_P(0) = 2B(P),$$

with $H_P(w) = O(1)$ uniformly on the required contour for the seven-prime family, so

$$J_P(z) = 2B(P) + O_A((\log z)^{-A})$$

uniformly for $z = 2P \pm 1$. This is one full- q principal identity, not a separate main term for each dyadic q cell.

Exact SmallQ/SmallR/Hard partition. With $U = X^{1/3}$, split $qr = 2P + s$ into

$$q \leq U, \quad q > U, \quad r \leq U, \quad q > U, \quad r > U.$$

The sectors are exhaustive and disjoint at the sharp level; the Hard sector is subsequently dyadically smoothed.

SmallQ hostile-audited pass. The former direct Gate-0 Type-I dictionary is superseded as a controlling provider because q divides $2P + s$, not P . The replacement is a primitive-conductor split $3 + 4$ multiplicative large sieve. The literal three- and four-prime coefficients satisfy

$$\sum |a_3|^2 \ll X^{3/7}(\log X)^{-3+o(1)}, \quad \sum |b_4|^2 \ll X^{4/7}(\log X)^{-4+o(1)}.$$

For squarefree $q \asymp Q$, the conductor split has the form

$$L_{Q,D}(c) \ll (\log X)^{O(1)} \left(Q + \frac{N}{D} \right) \|c\|_2^2,$$

because $q = fk \asymp Q$, $f > D$ implies $k \ll Q/D$. The low primitive-conductor sector with $D = (\log X)^{30}$ is routed through Siegel–Walfisz; the high-conductor sector also passes. Hence

AFFINE287-SP2-SMALLQ-34LS-NORMALIZATION45: HOSTILE-AUDITED PASS.

SmallR owner and general-modulus conductor split. For arbitrary modulus r , if χ is induced from primitive $\chi^* \bmod f$ and $j = r/f$, the exact relation is

$$\chi_r(n) = \chi^*(n) \mathbf{1}_{(n,j)=1},$$

with no assumption that r is squarefree or $(f, j) = 1$. Also

$$\phi(fj) \geq \phi(f)\phi(j), \quad f > D, \quad fj \asymp R \implies j \ll R/D,$$

so the general-modulus conductor estimate

$$L_{\text{gen}}(R, D; c) \ll (\log X)^{O(1)} \left(R + \frac{N}{D} \right) \|c\|_2^2$$

survives.

Let

$$D_{\text{sr},s} = S_{\text{sr},s} - M_{\text{sr},s}^{\text{prin}}.$$

The principal restriction is owned by the one global Euler packet and satisfies

$$M_{\text{sr},s}^{\text{prin}} \ll X^{1/3+o(1)},$$

while the audited direct SmallR source obeys

$$S_{\text{sr},s} \ll X(\log X)^{-16.5+o(1)}.$$

Thus

$$D_{\text{sr},s} = o(X/\log X)$$

and

AFFINE287-SP2-SMALLR-OWNER-SUBTRACTION45: HOSTILE-AUDITED PASS.

Hard physical range. For

$$q \asymp Q = X^\theta, \quad \frac{1}{3} < \theta < \frac{2}{3}, \quad M = X^{2/7}, \quad W_5 = X^{5/7},$$

put

$$T = \frac{W_5}{Q} = X^{5/7-\theta}, \quad z = T^{1/20}.$$

The physical fixed margin is

$$\delta = \frac{5}{7} - \frac{2}{3} = \frac{1}{21}, \quad \log z \geq \frac{1}{420} \log X.$$

For primes $\ell < z$, if $\ell \nmid q$ there is one forbidden t class modulo ℓ ; if $\ell \mid q$, there is no forbidden class because $\ell < z < Y$ cannot divide the physical w' . Hence the short- t upper sieve gives

$$\#\{t \text{ admissible}\} \ll \frac{T}{\log X} \frac{q}{\phi(q)}.$$

The corresponding Shiu progression map has modulus exponent $5/7$ below the admissible $3/4$ threshold by

$$\frac{3}{4} - \frac{5}{7} = \frac{1}{28}.$$

The corrected variance ledger is

$$V_Q^{RR} \ll \frac{MW_5^2}{Q} (\log X)^{-5+o(1)},$$

not $(\log X)^{-10}$; the second copy is consumed by the t -sieve. Moreover

$$V_Q^{PP} \ll \frac{MW_5^2}{Q} (\log X)^{-10+o(1)}, \quad V_Q^{RP} \ll \frac{MW_5^2}{Q} (\log X)^{-15/2+o(1)}.$$

Thus the raw-raw channel dominates:

$$V_Q \ll \frac{MW_5^2}{Q} (\log X)^{-5+o(1)}.$$

With the literal source cost $C_{\text{ext}} = 1$,

$$|D_{Q,s}| \ll X (\log X)^{-5/2+o(1)},$$

and summing $O(\log X)$ hard cells gives

$$\sum_Q |D_{Q,s}| \ll X (\log X)^{-3/2+o(1)} = o(X/\log X).$$

Hence

$$\boxed{\text{HARD-THETA-PHYSICAL-DELTA45: HOSTILE-AUDITED PASS, } \delta = 1/21.}$$

Exact full- q ownership and Balanced7 status. Use exactly one global Euler principal packet:

packet	owner
SmallQ principal	full Euler principal
SmallQ defect	3 + 4 conductor-split large sieve
SmallR principal	full Euler principal
SmallR defect	direct SmallR estimate
Hard principal	full Euler principal
Hard defect	short- t /Shiu physical-range estimate
even/nonunit	zero or explicit router

with $q = X^{1/3}$ owned by SmallQ and $r = X^{1/3}$ by SmallR. Even q is impossible for $2P + s$ odd; nonunit pieces are routed from the exact affine source. Therefore the principal restrictions sum to one and only one Euler principal packet:

$$\boxed{\text{ALL-Q-NO-DOUBLE-SPENDING45: HOSTILE-AUDITED PASS},}$$

$$\boxed{\text{ALL-Q-PROVIDER-REASSEMBLY45: HOSTILE-AUDITED PASS}.}$$

At the current analytic research/paper level,

$$\boxed{\text{SP2-BALANCED7-FULL-Q45: PASS}}$$

BALANCED7: HOSTILE-AUDITED RESEARCH/PAPER PASS.

The firewall is source provenance: the completed audit treats the literal SP-2 source, $H_*(P) = -20$, and seven prime-box source as supplied. The public manuscript repository does not presently expose the later 8213-job source-seal bank. Therefore this note states the Balanced7 verdict as conditional on the exact literal SP-2 source seal rather than claiming public-Lean source completeness.

Balanced7 effectivity remains open. The SmallQ low-conductor argument uses classical Siegel–Walfisz for $f \leq (\log X)^{30}$, whose threshold is generally ineffective. A separate lane remains

287-EFFECTIVE-LOWCOND-EXCEPTIONAL-PNT45: OPEN,

requiring an effective exceptional-zero-aware PNT/AP statement matching the project’s principal and exceptional comparison convention. No computable M_0 is claimed.

13.2 Uniform $k = 0$: the next source parent

Balanced7 does not complete the $k = 0$ fixed-certificate source. The next parent is

287-K0-SP2-UNIFORM-FRAGMENTATION-REASSEMBLY45: OPEN.

It must handle the actual k -range, coefficients

$$H_k = (-1)^r \binom{k-1}{r}, \quad r = \lfloor k\gamma_* \rfloor,$$

uniform analytic constants, fragmentation/polytopal multiplicity, squareful and repeated-prime strata, model-containing states and an exhaustive unique-owner map.

Earlier source work proposed a three-small-prime-prefix child. The latest determinant-one audit has moved the exact main-line residual further and more structurally.

13.3 Retracting the old large-shared- g_0 and near-frequency global closures

The local reciprocal calculations remain valid:

SHAREDGO-BPAIR-AVERAGED45: PASS, SHAREDGO-CAUCHY-CONFIGURATION45: PASS.

At Gram level the near-density gain is density $^{1/2}$ and after the final outer square root it is density $^{1/4}$. However the former global claims

DET1-LARGESHAREDGO-CELLS45: CLOSED, DET1-PRIMITIVE-NEARFREQ45: CLOSED

are retracted. Restoring the cross-level pair multiplicity of order G^2/H , the normalization $1/(g_1g_2)$, the first friable Cauchy A -factor and the reciprocal energies leaves a signed cross-level problem. Thus fixed-pair cancellation does not by itself close the full family.

13.4 Short-lift uniformity repair

For

$$m_{g,b}(\Delta) = \sum_{(d,2bg)=1} \frac{\mu(d)}{d} \Psi(d/\Delta),$$

the finite Euler factor is

$$H_{\mathfrak{H}}(w) = \prod_{p|\mathfrak{H}} (1 - p^{-1-w})^{-1}.$$

The corrected contour is

$$\Re w = -\frac{c_0}{\sqrt{\log \Delta}},$$

not the earlier t -dependent contour. For

$$\Delta \geq D_{\sharp} = \exp(C_{\sharp}(\log \log X)^2)$$

and $\mathfrak{H} \leq X^C$, the extremal prime-divisor argument gives

$$H_{\mathfrak{H}}(w) \leq (\log X)^{o(1)}.$$

Hence $\exp(-c\sqrt{\log \Delta})$ defeats any fixed power of $\log X$ after choosing $C_{\sharp}$ large. The current research status is

DET1-SHORTLIFT-EULER-COLLAPSE45: PASS AFTER UNIFORMITY REPAIR.

The analytic zero-free-region input remains uninhabited in Lean; no formal zeta zero-free theorem is inferred.

13.5 Product-modulus sign compression and fixed- D rigidity

Write

$$g_1 = g_0 r_1, \quad g_2 = g_0 r_2, \quad (r_1, r_2) = 1, \quad (g_0, r_1 r_2) = 1,$$

in the squarefree clean sector, and set

$$n = r_1 r_2.$$

Then exactly

$$\boxed{\mu(g_1)\mu(g_2) = \mu(r_1)\mu(r_2) = \mu(n),} \quad [g_1, g_2] = g_0 n.$$

Thus two level Möbius signs compress to one. If

$$D = t_1 r_2 - t_2 r_1,$$

then $(D, n) = 1$ in the primitive-frequency source, and for fixed D

$$t_1 \equiv D \bar{r}_2 \pmod{r_1}, \quad t_2 \equiv -D \bar{r}_1 \pmod{r_2}.$$

The common residual freedom is the shared- g_0 parameter, while

$$\boxed{\frac{t_1}{g_1} - \frac{t_2}{g_2} = \frac{D}{g_0 n}.$$

Consequently the friable kernel is

$$\Phi_A\left(\frac{D}{g_0 n}\right),$$

independent of the divisor split $r_1 \mid n$.

The exact research labels are

DET1-LEVELPAIR-PRODUCTMOD-SIGN45: PASS

DET1-FIXEDD-FREQUENCY-RIGIDITY45: PASS.

These are latest research-exact reductions; their formal banking is pending in the public manuscript repository.

13.6 CRT reciprocal normal form and the fixed- n firewall

After separating the shared- g_0 component, CRT combines the residual reciprocal phases modulo r and n/r . If

$$\beta_r \equiv b_1 \pmod{r}, \quad \beta_r \equiv b_2 \pmod{n/r},$$

the remaining phase has the exact form

$$e_n(sD \overline{2g_0\beta_r}).$$

Thus

$$\boxed{\text{DET1-LEVELPAIR-N-RECIPROCAL-NORMALFORM45: PASS}}$$

and the divisor-split kernel is a squarefree unitary-divisor additive-reciprocal/Kloosterman-idempotent object, not an ordinary Dirichlet character.

For fixed squarefree n , each prime $p \mid n$ has exactly two states: assigned to r or n/r . After Mellin separation of balanced support, the divisor split factors over $p \mid n$. But the local factor contains $\overline{n/p} \pmod{p}$; it therefore depends on the complementary coprime component. Hence

$$\boxed{\text{DET1-LEVELPAIR-PRIMEASSIGNMENT45: PASS AT FIXED } n},$$

but

$$\boxed{\text{DET1-LEVELPAIR-SINGLEMOBIUS-N45: OPEN}},$$

and no ordinary Euler-product/ $1/\zeta$ cancellation follows from the single $\mu(n)$ compression.

13.7 Shared-gcd Gram-as-square: the main structural reduction

Let $\Omega_H((g_1, g_2))$ select the shared-gcd H sector and define its divisor-lattice Möbius transform

$$\lambda_H(d) = \sum_{e \mid d} \mu(d/e) \Omega_H(e) = (\mu * \Omega_H)(d).$$

Exact Möbius inversion gives

$$\Omega_H((g_1, g_2)) = \sum_{\substack{d \mid g_1 \\ d \mid g_2}} \lambda_H(d).$$

Consequently the signed cross-level Gram reorganizes exactly as

$$Q_H = \sum_d \lambda_H(d) \sum_a \rho(a) \left| \sum_{d \mid g} \frac{\mu(g)}{g} V_g(a) \right|^2.$$

In the squarefree clean sector write $g = dm$, $(d, m) = 1$. Since $\mu(g) = \mu(d)\mu(m)$ and the outer $\mu(d)$ disappears in the square,

$$Q_H = \sum_d \frac{\lambda_H(d)}{d^2} \sum_a \rho(a) \left| \sum_{\substack{m \\ (m,d)=1}} \frac{\mu(m)}{m} V_{dm}(a) \right|^2.$$

At research level the harmonic projector satisfies

$$\sum_d \frac{|\lambda_H(d)|}{d^2} \ll \frac{1}{H}$$

on the stated H support. Thus the former two-level G^2/H signed-pair combinatorics is replaced by harmonic shared-gcd mass $1/H$ and a one-dimensional signed m family.

This is the principal new exact structural result:

$$\boxed{\text{DET1-SHAREDGCD-GRAM-SQUARE45: PASS EXACT}},$$

with public formal banking still pending.

13.8 Provisional one-level energy subpolytope

A coefficient-blind calculation gives the research bound

$$|Q_H| \ll \frac{(A+G)(GB+B^2)}{H} (\log X)^{o(1)}.$$

Against the required scale

$$AB^2(\log X)^{-6-2\eta-C_{\text{route}}},$$

it gives fixed-power saving on the strict subpolytope

$$\kappa > \max(\alpha, \theta) + \max(\beta, \theta) - 1 + \delta,$$

where

$$A = X^\alpha, \quad B = X^\beta, \quad G = X^\theta, \quad H = X^\kappa, \quad \alpha + \beta = 1.$$

The status is deliberately conservative:

$$\begin{array}{c} \text{DET1-SHAREDGCD-ONELEVEL-ENERGY45} \\ \text{PROVISIONAL RESEARCH CLOSURE ON A STRICT SUBPOLYTOPE} \end{array}$$

A hostile promotion audit is still required; this is not a publication theorem and does not alter the global OPEN status.

13.9 Why complete primitive- t Ramanujan completion is nonclosing

Fully completing the primitive-frequency variable gives schematically

$$V_g(a) = \sum_b \beta_g(b) c_g(2ab + s).$$

For squarefree m and $N = 2ab + s$,

$$p \nmid N : \quad c_p(N) = -1, \quad \mu(p) c_p(N) = +1,$$

while

$$p \mid N : \quad c_p(N) = p - 1, \quad \mu(p) c_p(N) = -(p - 1).$$

Thus the generic Ramanujan local sign cancels the remaining Möbius sign. The formal Euler object contains

$$\prod_p (1 + p^{-1-w}) = \frac{\zeta(1+w)}{\zeta(2+2w)},$$

which has a pole at $w = 0$, not a $1/\zeta$ zero. Therefore

$$\text{ONELEVEL-RAMANUJAN-ZEROFREE45: REPRESENTATION LOOP / NONCLOSING.}$$

The remaining sign must be used before complete primitive-frequency Ramanujan reassembly neutralizes it. This is a source-preservation issue, not a generic fixed-Chowla obstruction.

13.10 Current first exact main-line residual

For the current shared-gcd sector define the one-level object

$$\mathcal{G}_{H,d} = \sum_{a \asymp A} \rho(a) \left| \sum_{\substack{m \asymp G/d \\ (m,d)=1}} \frac{\mu(m)}{m} \sum_{t \bmod dm}^* \tilde{G}_{dm,s,\tau}(t) e_{dm}(ta) \right|^2.$$

The required estimate is

$$\left| \sum_d \frac{\lambda_H(d)}{d^2} \mathcal{G}_{H,d} \right| \ll AB^2 (\log X)^{-6-2\eta-C_{\text{route}}}.$$

The current generic bound is only

$$\frac{(A+G)(GB+B^2)}{H} (\log X)^{o(1)}.$$

Therefore the first exact main-line analytic residual is

$$287\text{-K0-SP2-DET1-SHAREDGCD-ONELEVEL-MOBIUS-GRAM45}.$$

The missing mechanism is cancellation in

$$\sum_m \frac{\mu(m)}{m} V_{dm}(a)$$

while preserving the primitive t variable, reciprocal b source, friable a average and exact shared-gcd projector. Complete primitive- t Ramanujan completion is explicitly excluded as the cancellation mechanism because it neutralizes the sign.

Conceptual lesson. The successive reductions expose a recurring source-preservation phenomenon. Local reciprocal cancellation is insufficient after the full cross-level multiplicity is restored. The successful reorganization instead keeps the Möbius signs until the shared-gcd Gram is converted exactly into a one-level signed operator. Conversely, complete primitive-frequency Ramanujan reassembly neutralizes the remaining Möbius sign. The present frontier is therefore a source-specific one-level Möbius–primitive-frequency dispersion problem, not a generic two-point Chowla problem.

14 Verification manual: what is actually proved

14.1 Table 1: public and supplied machine-checked objects

Object	Verification status	Exact scope / nonclaim
Finite #287 theorem through 4×10^9	PUBLIC LEAN-CHECKED	Exact finite exclusion only; no large- M theorem.
WindowPairSupply compiler	PUBLIC LEAN-CHECKED / CONDITIONAL	Compiler from eventual effective supply; supply itself unproved.
V15 $\mu * \log$ / determinant-one algebra	PUBLIC-MAIN SYNCHRONIZED	Finite/source algebra; no analytic cancellation.
V16 factorial Euler / local Euler / linearity	PUBLIC-MAIN SYNCHRONIZED	Factorial polarization and formal Euler algebra; no analytic endpoint theorem.
V17–V19 3221 finite/source banks	PUBLIC-MAIN SYNCHRONIZED	Grouping, finite energy, source reconstruction, pre-Poisson and character-Gram infrastructure; analytic inputs remain interfaces.
Primitive Ramanujan / short-lift / shared- g_0 finite routers	SUPPLIED FORMAL BANK	Finite algebra/reassembly and local counting only; analytic zero-free inputs are uninhabited.
Post-Balanced7 8213-job safe bank	SUPPLIED LEAN SAFE BANK / REPOSITORY SYNC PENDING	Reported modules include seven-box weights, conductor-split LS compiler, small-prime prefix, owner and uniform-fragmentation compilers; current public manuscript repository does not contain these modules.
Product-modulus / fixed- D / CRT / Gram-square layer	RESEARCH-EXACT / FORMAL BANKING PENDING	Do not call public-Lean checked until a successful bank is supplied and synchronized.

14.2 Table 2: externally audited analytic inputs

Input	Status	Scope
Balanced7 SmallQ 3 + 4 conductor-split large sieve	HOSTILE-AUDITED PASS	Includes the literal L^2 norms and low/high primitive-conductor split.
General-modulus SmallR conductor large sieve	HOSTILE-AUDITED PASS	Uses $\chi_r = \chi^* \mathbf{1}_{(n,j)=1}$ with no squarefree/coprime cofactor assumption.
Short- t one-dimensional upper sieve	HOSTILE-AUDITED PASS	Physical hard range only; standard source-specific upper-sieve application.
Shiu divisor average	HOSTILE-AUDITED PASS	Progression modulus exponent 5/7 has margin 1/28 below the audited 3/4 threshold.
Balanced7 full- q analytic reassembly	HOSTILE-AUDITED RESEARCH/PAPER PASS	External analytic verdict; conditional on exact SP-2 source seal, not a Lean analytic proof.
Short-lift zero-free-region collapse	RESEARCH/NANC PASS AFTER UNIFORMITY REPAIR	The zero-free analytic statement is not Lean-formalized.

14.3 Table 3: open source/provenance identifications

Source obligation	Status	Meaning
Exact SP-2 seven-box source seal	SOURCE-OPEN IN PUBLIC MANUSCRIPT REPO	The hostile Balanced7 audit treats the physical seven-box source as supplied; later 8213-job formal bank is not public-synchronized here.
Gate-1A five physical pins / SB- ν source glue	SOURCE-OPEN	Required only for packets actually censused into the canonical common-weight provider.
Gate-1B comparison/main-term and repaired top-band dictionary	SOURCE/ANALYTIC OPEN	Independent from the present #287 uniform- $k = 0$ frontier.
Ford generated-(7.23) census / SOURCE-ADAPTER45 / C_{FM}	OPEN	Needed only for proof-specific replacements that actually use the Ford generated family.
Balanced R_9 literal $G(d; n)$ specialization	SOURCE-OPEN	Finite 70 identity does not by itself give physical $H(n) = 70$.
Effective low-conductor exceptional PNT dictionary	OPEN	Needed for computable threshold/effectivity; classical Siegel–Walfisz is insufficient.

14.4 Table 4: current open analytic theorems

Theorem / node	Exact task	What remains if proved
287-K0-SP2-DET1-SHAREDGCD-ONELEVEL-MOBIUS-GRAM45	Obtain the required logarithmic cancellation in the shared-gcd one-level signed Möbius Gram while retaining primitive t , reciprocal b , friable a and exact projector.	Uniform $k = 0$ reassembly/census, remaining fixed- g_* packets, FCL, N_2 , Gate 2, effectivity and eventual supply.
DET1-SHAREDGCD-ONELEVEL-ENERGY45	Promote or refute the provisional strict-subpolytope coefficient-blind closure.	Even a positive audit does not automatically cover the full polytope.
Gate-1B C4SHIFT-BEZOUT-DUAL-ROELL45	Prove source-specific cancellation in the coupled nonzero (ν, r_0, ℓ) Bézout/gamma dual family; audit the feeding D -census.	Zero-mode local match, upstream C4Shift promotion, Pure5/defect strata, QK56 and Gate 1B remain.
FCL / source-specific N_2	Control all remaining fixed-certificate leakage and exceptional collars.	Gate 2 and effectivity.
287-EFFECTIVE-LOWCOND-EXCEPTIONAL-PNT45	Replace ineffective low-conductor input with an effective exceptional-zero-aware theorem matching the comparison convention.	Effective Gate 2 and eventual WindowPairSupply still require the remaining source programme.
Eventual WindowPairSupply	Produce the effective adjacent prime-power-window witness for all sufficiently large M .	Existing finite Lean compiler then proves the public statement.

15 Current numbered route to Erdős #287

The current dependency route is:

1. source-seal the exact SP-2 Balanced7 seven-box adapter in the public formal record;

2. retain the hostile-audited Balanced7 research/paper closure, while keeping its effective version separate;
3. close 287-K0-SP2-DET1-SHAREDGCD-ONELEVEL-MOBIUS-GRAM45;
4. complete 287-K0-SP2-UNIFORM-FRAGMENTATION-REASSEMBLY45, including all actual k values, multiplicities, repeated/squareful strata and unique owners;
5. complete the remaining fixed- g_* packet census and census-driven provider assignment;
6. close fixed-certificate leakage (FCL);
7. close the source-specific N_2 collar;
8. activate Gate 2 at one compatible ε_* ;
9. discharge the separate effective low-conductor/exceptional PNT lane and make positive witness extraction computable;
10. prove eventual WindowPairSupply, or a stronger effective witness implying it;
11. invoke the existing finite Lean compiler.

Later results may collapse several steps. In particular, the first analytic residual above is a statement about the current determinant-one shared-gcd sector; it is not claimed to be an irreducible obstacle of every possible formulation.

The non-implications remain:

Balanced7 research closure $\nRightarrow$ uniform $k = 0$,
 uniform $k = 0 \nRightarrow$ FCL without packet exhaustiveness,
 FCL $\nRightarrow$ effective Gate 2 or WindowPairSupply,
 no local packet estimate alone proves Erdős #287.

16 How an external reviewer can falsify or improve the programme

A serious review is most useful if it attacks one of the following:

1. a source dictionary that changes the physical packet;
2. a theorem-slot mismatch between an actual generated packet and a claimed provider;
3. a missing normalization or prefactor;
4. a multiplicity or support-enlargement loss;
5. double-spending of a Möbius sign or analytic saving;
6. an incorrect comparison/principal/exceptional-character main term;
7. an incomplete Ford-generated packet census;
8. an invalid invocation-by-invocation replacement of Ford–Maynard’s universal Type II;
9. an endpoint/effectivity gap;
10. a mismatch between a claimed Lean status and the actual kernel/public bank.

Failure of any conditional provider does *not* invalidate the machine-checked finite theorem through 4×10^9 ; it invalidates only the corresponding large- M route.

17 Glossary of recurring programme terms

Gate 0	The physical #287 source/comparison/Type-I package.
Gate 1A	Canonical common-weight moving-source Type-II provider (G1A-source).
Gate 1B	Complementary signed/Möbius/high-order Type-II provider based on (G1B-center)–(G1B-M).
Gate 2	Source-specific Ford–Maynard lower-bound compiler.
BPP	Band-limited prime-participation mechanism converting moving-prime participation to a family-energy saving in Gate 1A.
SB-ν	Synchronized two-view/source-glue fibre condition in the Gate-1A source adapter.
Pure5	Five-centred-defect distinguished-prime source family in the Gate-1B chain.
NearPrim	Near-primitive/full-conductor slice ($r = 1$) of the Gate-1B conductor family.
QK56	Full same-/cross-modulus covariance parent (QK56-parent).
SOURCE-ADAPTER45	Literal source dictionary from actual Ford-generated packets to proved providers.
C_{FM}	Invocation-by-invocation certificate replacing the universal Type-II hypothesis in the chosen Ford proof by generated-packet estimates.
R_9	Balanced nine-coordinate Ford-generated leakage candidate; finite binomial coefficient 70 is proved, physical $G(d; n)$ specialization is not.
FCL	Fixed-certificate leakage/correlation package (FCL).
N_2	Exceptional Ford collar requiring a separate source-specific aggregate bound.
Balanced7	Fixed-degree seven-prime-box V-cell at $Y = X^{1/7}$.
V-cell	A generated packet in the model/factorial seven-prime branch.
Möbius singleton	Generated short Type-II factor carrying the Möbius sign.
fixed-g_*	One fixed positive Ford certificate used in the leakage transference.
WindowPairSupply	The exact large- M adjacent prime-power window predicate consumed by the finite Lean compiler.

18 The source-minimal affine algebra

The current clean narrative starts from the exact identity $\Lambda = \mu * \log$. The supplied V15 Aristotle replay report records a successful 8118-job kernel build of the declarations summarized below. Because public repository **main** has not yet synchronized that replay, their status in this candidate is **LEAN-PROVED IN LATEST V15 REPLAY**, not **PUBLIC LEAN-CHECKED**. No analytic estimate is encoded by these identities.

18.1 The identity $\Lambda = \mu * \log$

For the genuine arithmetic functions,

$$\mu * \log = \mu * (\Lambda * 1) = (\mu * 1) * \Lambda = \Lambda,$$

and hence

$$\Lambda(N) = \sum_{qr=N} \mu(q) \log r.$$

At the affine value $N = 2mn + s$, $s \in \{-1, +1\}$,

$$\Lambda(2mn + s) = \sum_{qr=2mn+s} \mu(q) \log r.$$

The V15 replay also preserves the natural-number sign firewall for the minus case.

18.2 Exact factor partition and determinant-one line

For a cutoff U , factor pairs split disjointly into

$$q \leq U, \quad q > U, r \leq U, \quad q > U, r > U.$$

The hard class is exact rather than heuristic:

$$\sum_{\substack{qr=2mn+s \\ q>U, r>U}} \mu(q) \log r.$$

Every hard pair satisfies

$$qr - 2mn = s, \quad s = \pm 1.$$

This implies

$$(r, 2m) = 1, \quad (q, 2n) = 1.$$

Fixing r, m and one solution (q_0, n_0) , all integer solutions are

$$q = q_0 + 2mt, \quad n = n_0 + rt, \quad t \in \mathbb{Z},$$

with unique t . The V15 replay report records nonvanishing, the coprimality statements, both directions of this parametrization, and multiplicity-one uniqueness as kernel-checked.

The same replay proves only the finite exponent ledger, not an asymptotic estimate. In particular,

$$\frac{1}{6} - \frac{16623}{100000} = \frac{131}{300000} > 0.$$

19 The balanced-seven factorial Euler reduction

The factorial Euler family remains the clean algebraic encoding:

$$a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p), \quad F_{\mathbf{z}}(p^e) = \frac{a_{\mathbf{z}}(p)^e}{e!}.$$

For $\Omega(n) = 7$,

$$7^7 [z_1 \cdots z_7] F_{\mathbf{z}}(n) = \sum_{\substack{p_1 \cdots p_7 = n \\ p_i \text{ ordered}}} \prod_{i=1}^7 \omega_i(p_i),$$

including repeated primes exactly. Coefficient extraction commutes with every linear expected-term operator.

The controlling physical source is nevertheless the fixed-certificate SP-2 packet, not a putative unique Ford-(7.23) generated packet. The completed hostile audit gives the research/paper-level Balanced7 full- q pass described in Section 13: SmallQ, SmallR and Hard defects are controlled with one global Euler principal owner. The public-formal source seal of the literal seven-box SP-2 adapter is still pending, and the effective low-conductor version remains open.

Therefore the old sequence of first residuals

3221-DI-KUZNETSOV-LITERAL-SPLICE45,
3221-HIGHQUOTIENT-FIVEBOX-SHIFTED-GRAM45,
BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45

is provenance only. The current main line has moved to uniform $k = 0$ and the one-level Möbius Gram of Section 13.

20 Published distribution results and the exact dictionary audit

Drappeau–Granville–Shao work with the class $\mathcal{C}$ of *multiplicative* functions satisfying $|\Lambda_f(n)| \leq \Lambda(n)$; complete multiplicativity is only a special case. Their theorem reaches moduli $x^{3/5-\varepsilon}$ for sufficiently smooth support, so it does not literally supply the exact endpoint $Q = X^{3/5}$ required above.

For Pascadi’s current arXiv v2, the relevant stated result is the smooth-number theorem (Theorem 1.5, refined as Theorem 6.4). Immediately after Theorem 1.5 the paper remarks that a similar result for smooth-supported multiplicative functions can be deduced by a slight extension of Proposition 6.3; the current v2 does not state a separate “Theorem 1.7” giving the black box required here. More importantly, the written proof architecture itself is quantitatively nonclosing at the seven-prime smoothness scale.

Indeed, renaming the proof’s internal small parameter as η , the proof arranges

$$Q \leq x^{5/8-100\eta}, \quad y \leq x^\eta.$$

At the required modulus

$$Q = x^{3/5},$$

the first inequality implies

$$\frac{3}{5} \leq \frac{5}{8} - 100\eta \implies \boxed{\eta \leq \frac{1}{4000}}.$$

But the source has

$$y = x^{1/7},$$

so the second inequality would require

$$\boxed{\eta \geq \frac{1}{7}}.$$

These conditions are incompatible. Thus the *direct existing Pascadi proof route*, with its written parameters, does not furnish the factorial endpoint theorem at $(Q, y) = (X^{3/5}, X^{1/7})$. This is a theorem-dictionary/proof-parameter audit, not a claim that no future extension of Pascadi’s technology can succeed.

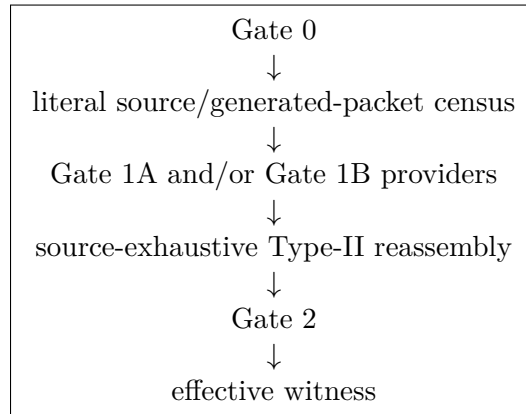
This explains the current research direction: preserve the literal fixed degree seven rather than forcing a generic smooth-factorization parameter, carry that structure through dispersion, and test a post-dispersion four-block arrangement. The “fixed-degree-seven / four-block” programme is a research direction, not a proved theorem or an already matched Gate-1B provider. Its purpose is precisely to attack the narrower signed torus coefficient without paying for a theorem uniform over all smooth-supported functions.

The earlier V14 Vaughan decomposition remains mathematically valid as a nonminimal alternative source audit. It is no longer presented as a co-equal controlling branch in this public narrative; its prime-modulus two-outer packet and related failed dictionaries belong in the technical provenance appendix.

Current source-preservation lesson. The earlier smooth-number, DI/Kuznetsov, HHH and large-shared- g_0 lanes remain valuable audit history, but none is the controlling uniform- $k = 0$ frontier. The latest exact reorganization keeps the Möbius sign through the shared-gcd Gram-as-square reduction. Fully completing the primitive frequency is explicitly nonclosing because the Ramanujan local sign generically neutralizes μ . Thus the current target is a one-level signed Möbius-primitive-frequency dispersion theorem rather than another generic Kloosterman black box.

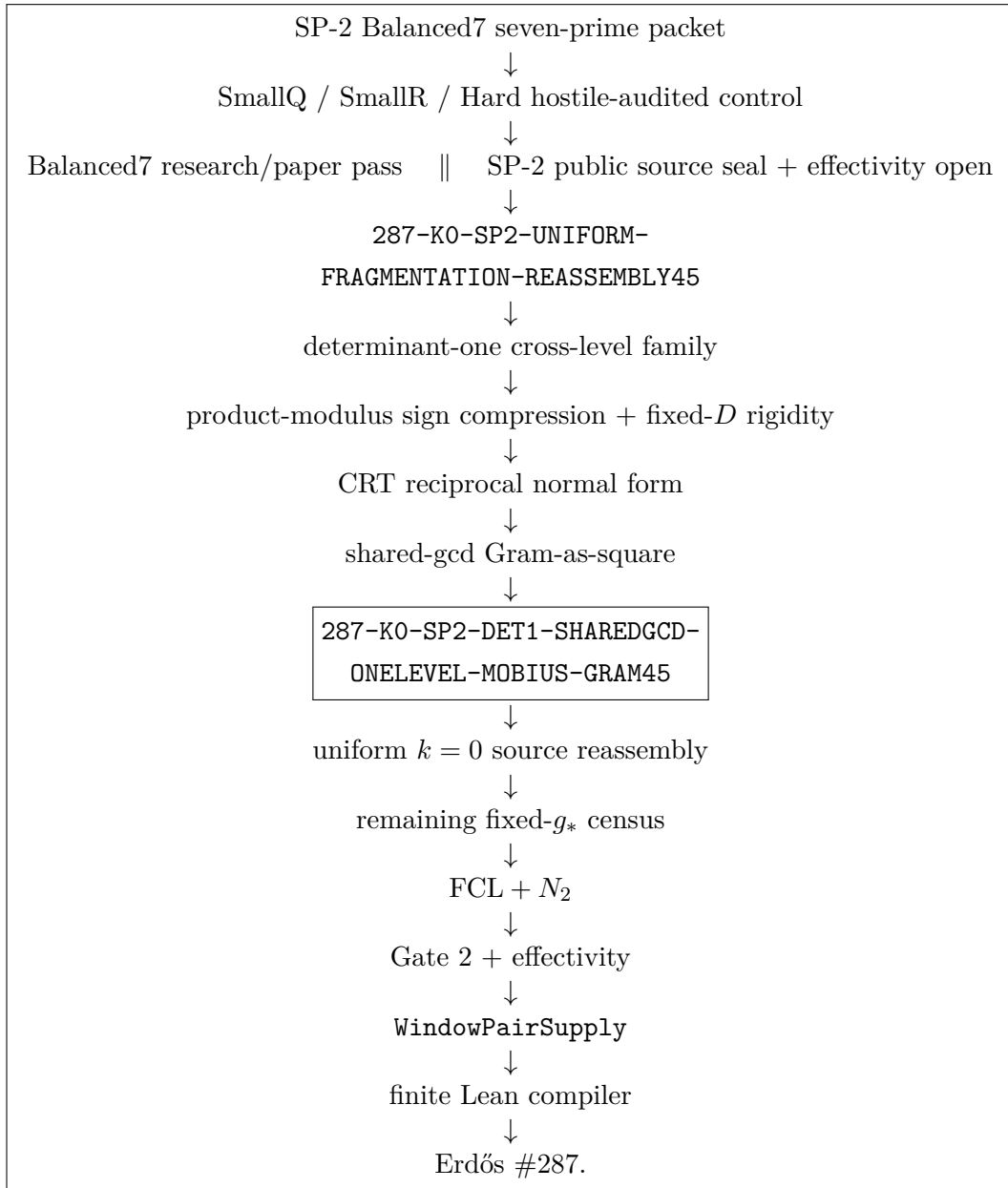
21 The current end-to-end dependency graph

The stable Gate spine is unchanged:



Provider assignment remains census-driven; the present #287 main line does not assume Gate 1A or Gate 1B globally.

The current #287 overlay is



Every arrow remains a firewall: a local research pass is not silently promoted to a source-exhaustive, effective or formal theorem.

22 Formal verification and reproducibility

22.1 Public manuscript repository

The public aggregate formal build presently imports the V15–V19 banks. Their supplied reports record successful builds of 8118, 8123, 8131, 8135 and 8140 jobs, respectively, with the stated zero-error and no-user-axiom firewalls. These banks cover, among other things, $\Lambda = \mu * \log$ source algebra, factorial Euler polarization, labelled 3221 regrouping, finite diagonal/off-diagonal/no-wrap compilers, pre-Poisson divisibility and high-conductor character-Gram infrastructure. They do not prove the external analytic estimates used by the hostile research audits.

22.2 Later supplied safe bank

A later supplied append-only safe-bank report records a post-Balanced7 status/axiom audit and modules

- `SevenBoxPrimeWeights`; `PrimeTupleMultiplicity`;
- `ConductorSplitLargeSieve`; `LowConductorSiegelWalfisz`;
- `SmallQ34LSCompiler`; `SmallPrimePrefix`;
- `PostRepairOwnerCompiler`; `UniformFragmentationCompiler`.

The reported build has 8213 jobs and zero errors. Those modules are not present in the current public manuscript repository at this audit cutoff. R7 therefore labels them

SUPPLIED LEAN SAFE BANK / REPOSITORY SYNC PENDING,

not PUBLIC-MAIN SYNCHRONIZED.

22.3 Determinant-one formalization generations

Already supplied/formally banked finite material includes primitive Ramanujan algebra and reassembly, the short-lift finite profile, harmonic twists, shared- g_0 u -parameter routing, averaged b -pair gcd identities, reduced-denominator algebra and exact near-frequency counting. Their analytic inputs remain uninhabited.

The newest product-modulus sign compression, fixed- D frequency rigidity, CRT n -reciprocal normal form and shared-gcd Gram-as-square are

latest research-exact layer / formal banking pending

unless and until a successful corresponding formal replay is supplied and synchronized. R7 does not infer Lean verification from research nomenclature.

22.4 External analytic firewall

The Balanced7 SmallQ large-sieve, SmallR estimate, one-dimensional upper sieve, Shiu divisor average and short-lift zero-free-region inputs are external analytic results or research-audited applications. A Lean compiler may prove implications from named inputs; it does not thereby formalize the external theorem. Likewise the hostile-audited Balanced7 research/paper pass is not labelled a Lean analytic theorem.

The first exact one-level Möbius Gram is presently an open research theorem; no interface is treated as inhabited merely because the underlying exact algebra has been reduced.

23 Conclusion

The strongest unconditional result remains the public machine-checked finite theorem: no exact counterexample exists with maximum denominator at most 4×10^9 . The exact global large- M interface remains `WindowPairSupply`; no eventual effective supply theorem is proved.

The Balanced7 fixed-certificate seven-prime packet has now passed the completed hostile analytic audit at the research/paper level. The direct source has $H_*(P) = -20$, the SmallQ conductor-split $3 + 4$ large-sieve route passes, the general-modulus SmallR owner is controlled, the physical hard range has the fixed margin $\delta = 1/21$, and the full- q principal/defect ownership reassembles without double spending. This verdict remains subject to the exact public SP-2 source seal and is not an effective or Lean analytic closure. In particular, the classical low-conductor step does not provide a computable threshold.

The controlling main line has therefore moved to uniform $k = 0$. Two former global closure claims are withdrawn: large shared- g_0 cells and primitive near-frequency cells are not globally closed

merely from their local reciprocal-density gains once the full cross-level G^2/H multiplicity and all normalizations are restored.

The latest exact reorganization is more structural. Product-modulus compression replaces the two level signs by one $\mu(n)$ sign; fixed- D rigidity freezes the primitive frequency residue classes; CRT gives a squarefree unitary-divisor reciprocal kernel; and the shared-gcd projector converts the signed cross-level Gram exactly into

$$Q_H = \sum_d \frac{\lambda_H(d)}{d^2} \sum_a \rho(a) \left| \sum_m \frac{\mu(m)}{m} V_{dm}(a) \right|^2.$$

The apparent two-level signed family is therefore a one-level Möbius operator. A coefficient-blind estimate presently closes only a strict subpolytope and is provisional. Complete primitive-frequency Ramanujan reassembly is nonclosing because it neutralizes the remaining Möbius sign.

Accordingly the historical R7 first exact main-line analytic residual was

$$\boxed{287-K0-SP2-DET1-SHAREDGCD-ONELEVEL-MOBIUS-GRAM45.}$$

Its task is to exploit the one-level Möbius sign while preserving primitive frequency, the reciprocal b source, the friable a average and the exact shared-gcd projector.

Uniform $k = 0$ remains open. FCL has not yet been reached; N_2 and Gate 2 are not activated; effectivity remains open; eventual WindowPairSupply remains open.

$$\boxed{\text{Erdős Problem \#287 remains open.}}$$

Part II

Current Frontier and Provenance Ledger

24 Purpose, coverage baseline and status firewall

Part II restores the *technical depth* of the earlier public dossier without restoring its chronological patch-stack format. Its coverage floor is the V15 public-review snapshot committed as

7fc3797105519c4e6a1669f174ce7394336bdc18,

which published the former 50-page-style Gate-1A/Gate-1B/Ford programme together with the then-current balanced-seven update. That snapshot is used here only as a coverage baseline: when later work changes the controlling route, the later status wins.

Three time/status layers are kept separate throughout:

1. **public/formal bank:** theorem or compiler actually present in the public Lean project;
2. **latest supplied research ledger:** a mathematically audited frontier supplied after the V15 snapshot but not silently promoted to a public Lean or published theorem;
3. **open/source interfaces:** deliberately unproved analytic or source-identification obligations.

This is important for Gate 1B. The public formal socket still records the one-completion coefficient

$$\beta_{D,P}(d, p) = \mu_D(d) \Lambda_P(p)$$

and the exact 1/104 distribution-margin budget. That socket is neither false nor erased. However, the latest supplied analytic work has moved beyond it to a Motohashi/rank-one frontier. The formal socket is therefore a *banked abstraction/alternative provider interface*, while the rank-one chain below is the current research frontier.

Detailed dead-route archaeology remains outside the main narrative. What is restored here is every current component needed to reconstruct the proof programme: Gate 1A source class, Gate 1B current frontier and downstream firewall, the Ford-generated packet census, balanced- R_9 /fixed-certificate leakage, Gate 2, and the relation of the current factorial- Ω_7 source to those providers.

25 Gate 1A: current canonical-provider ledger

Gate 1A is a conditional theorem for one canonical common-weight physical operator. It is not a universal theorem for arbitrary row-dependent source weights and it is not assumed globally necessary for Erdős #287.

For a row

$$e = (r, k, m), \quad m' = m + kr,$$

the affine forms satisfy

$$m' A_e(t) - m B_e(t) = 2k,$$

and the canonical physical operator is

$$C_e(b, d) = \sum_t W_D(t) \left(\sum_{p \preceq L} b_p \rho_p(A_e(t)) \right) \left(\sum_{q \preceq L} d_q \rho_q(B_e(t)) \right).$$

The conditional target remains

$$\sum_{e \in \mathcal{E}^\#} |C_e(b, d)|^2 \ll \frac{ML^4}{H} X_1^{o(1)} \sum_{p \preceq L} |b_p|^2 \sum_{q \preceq L} |d_q|^2,$$

with the same normalization conventions as in Part I.

25.1 What is banked and what is still external

Gate-1A component	Status	Current interpretation
Determinant/row algebra and all- m row family	BANKED FINITE/OPERATOR	The theorem is not restricted to clean- P_3 rows.
Canonical common physical weight W_D	SOURCE FIREWALL BANKED	Later row dependence must be deterministically derived from this one datum.
Scale ledger, row-energy conservation and normalization bridge	BANKED	Includes the H^2 conversion and the signed vertex margins.
Finite BPP participation-to-energy compiler	BANKED	Participation implies a moving-family energy saving once the physical participation input is supplied.
Projective ratio-class/axis algebra	BANKED FINITE ALGEBRA	The state diagonal is routed inside the zero-curvature/projective branch rather than discarded.
Outer-square-root algebra and exponent budget	BANKED	The formal compiler checks the square-root conversion but does not manufacture the physical energy pin.
Poisson–Bruhat transport/source ceiling	SOURCE PIN	Must preserve the literal physical packet and absolute comparator.
Fixed-state exclusion and prime participation	EXTERNAL PIN	The finite compiler is present; the physical smooth-envelope/short-interval input is external.
Outer-root physical energy identification	SOURCE OPEN	Identifies the abstract BPP energy with the actual Gate energy.
Projective/diagonal physical source routing	SOURCE OPEN	Finite algebra exists; the physical source ceiling must be supplied at target scale.
Literal recombination estimate	SOURCE OPEN	Controls discarded/excision/recombination errors without spending the same saving twice.
Arbitrary edge-dependent D_2 family	OUTSIDE CANONICAL THEOREM	Requires an upstream Gate-0 adapter or finite-template/nuclear reduction.

The current source-glue issue can be summarized as a synchronized two-view/fibre problem: the same physical packet has to satisfy the common-weight representation, the BPP participation representation and the final source-routing representation simultaneously. We refer to this compactly as the *SB- ν /source-glue interface*. It is a source-identification problem, not an exponent deficit in the already-banked finite operator algebra.

25.2 When Gate 1A is required for #287

The correct logical statement is

GATE1A_REQUIRED is decided by the literal Ford-generated packet census.

An actual generated packet may use Gate 1A only after a scale/source dictionary supplies:

- the relation between the Gate ambient scale X_1 and the #287 scale X ;

- the identification of the physical bilinear variables with the Gate row variables;
- the induced R, L, H, K, U, D ranges;
- preservation of the one common physical weight, or a proved finite-template/nuclear reduction with acceptable total cost;
- survival of the binding fixed-power margin after truncation, multiplicity and scale conversion.

No global Gate-1A closure is assumed if the generated census never routes a needed packet to this source class.

26 Gate 1B: banked socket and current Bézout-dual frontier

Gate 1B remains the complementary signed/parity-breaking provider. Its banked one-completion socket

$$\beta_{D,P}(d,p) = \mu_D(d)\Lambda_P(p)$$

and the exact $1/104$ distribution-level capacity ledger remain valid, but they are not the current first research residual.

The historical progression through the Motohashi/fivefold source, rank-one endpoint and weighted shifted top-band repair is retained as provenance. The current public research index instead records the strict reduction

$$\text{C4SHIFT-OFFDIAG-CENTERED-AP5/8-GRAM45} \implies \boxed{\text{C4SHIFT-BEZOUT-DUAL-ROELL45}}.$$

On $(d, \ell) = 1$, $r = dr_0$, the Bézout line

$$bZ_2 - aZ_1 = \ell r_0$$

separates as

$$Z_1 = \ell r_0 \kappa_{a,b} + bt, \quad Z_2 = \ell r_0 \bar{b}_a + at,$$

with physical residue

$$t \equiv -2\overline{usdab} \pmod{\ell}$$

independent of r_0 . The centred cyclic dual kernel is

$$\widehat{K}_{\text{cent}}(\nu) = \ell^{-1} e_\ell(-\nu\tau) \mathbf{q}(\nu), \quad \mathbf{q}(\nu) = 1 - \ell^{-1} \sum_k \mathcal{M}_1(k + \nu \overline{dab}) \overline{\mathcal{M}_2(k)}.$$

The zero mode is a local owner and feeds `C4SHIFT-BEZOUT-ZEROMODE-LOCALMATCH45`; it is not estimated away. For $\nu \neq 0$, after opening the gamma source,

$$bx_2y_2 - ax_1y_1 = \ell r_0, \quad e_\ell(\nu t) = e_\ell(\nu \bar{b} x_1 y_1).$$

The provisional local census bound

$$\mathcal{N}_D \ll Y^{3/4} \sqrt{\min(D, T/D)} L^C, \quad T = Y^{3/2} L^{O(1)},$$

is `HOSTILE-AUDIT PENDING`. It suggests boundary-band closure and reduces the worst residual to $Y^{3/8} = X^{1/24}$, but the central D range remains and the claimed fixed-fibre uniqueness must be checked allowing t -differences by multiples of ℓ .

Accordingly

$$\boxed{\text{C4SHIFT-BEZOUT-DUAL-ROELL45}}$$

is the first exact Gate-1B analytic residual at research level. The missing mechanism is source-specific cancellation in the coupled nonzero (ν, r_0, ℓ) Bézout/gamma family. The parallel zero-mode/top-band local comparison is independently source-open.

26.1 Downstream Gate-1B firewall

Only after the nonzero Bézout child and the local zero-mode/top-band comparison both close may the compiler promote the centred-AP parent, one-minor pushed energy and Q-Fourier pushforward. These then feed the remaining beta/determinant/top-band descendants, Pure5 defect strata, NearPrim, the $r > 1$ recursion, QK56 exhaustiveness and final source reassembly. No earlier node is allowed to skip these descendants.

27 Ford–Maynard generated packets: census before provider

The Ford–Maynard universal Type-II hypothesis is a clean published sufficient wrapper, but Proposition 7.22 does not invoke arbitrary coefficient pairs out of thin air. After a finite decomposition it generates the grouped equation-(7.23) family

$$\sum_{n=n'n''\asymp x} Y_1(n')Y_2(n'')w_{n'n''}.$$

A proof-specific replacement is therefore legitimate only after two source obligations:

1. a *literal generated-packet census* listing every coefficient family produced by the proof;
2. a replacement certificate C_{FM} proving that every Type-II invocation is covered by one proved provider after all Mellin/Perron, support and norm transformations.

The programme labels

SOURCE-ADAPTER45,
FM-TO-GATE-COORDINATE-CENSUS45

refer to this source dictionary. Neither can be replaced by the slogan “the packet looks Type-II”.

27.1 Current provider-facing census

Generated family	Literal status	Candidate provider	Current blocker
Möbius singleton	source-specific generated packet	Gate 1B / signed source	literal dictionary and stability under generated outer factors
Model singleton	source-specific generated packet	QK56/direct affine completion	native packet dictionary; short Möbius carrier may sit in the complement
Balanced-seven V-cell	explicit fixed-degree source	factorial- Ω_7 endpoint theorem	signed endpoint + physical comparison identification
Balanced R_9 leakage	Ford-generated leakage candidate	fixed-certificate/FCL lane	literal $G(d; n)$ specialization and source reassembly
Common-weight/edge packets	census-dependent	Gate 1A only after exact match	scale/source adapter and common-weight preservation
Pure-five/high-order packets	Gate-1B source family	Motohashi/rank-one chain	endpoint off-diagonal, comparison, high- k , downstream strata
Remaining fixed- g_* packets	not exhaustively censused	TBD from literal packet shape	finite census and provider assignment

This table encodes the main logical correction:

provider assignment follows the generated source; it is not imposed globally.

In particular, Gate 1A is neither deleted nor assumed universally necessary.

28 Balanced R_9 , fixed-certificate leakage and the Gate-2 landing

The balanced order-nine identity

$$\sum_{j=0}^4 (-1)^j \binom{9}{j} = 70$$

is a genuine finite/binomial fact. The formal bank also records the complementary alternating identity. What is *not* automatically proved is that Ford's actual arithmetic weight $G(d; n)$ selects exactly this divisor-order specialization on the physical balanced- R_9 cell.

Therefore the correct status is:

balanced R_9 = Ford-generated leakage candidate,

with

$H(n) = 70$ only after the literal $G(d; n)$ specialization is proved.

The old R_9 death certificate and the abandoned unbalanced-1/5 redesign are not revived.

The fixed-certificate architecture uses one explicit positive Ford certificate g_* and its arithmetic weight

$$H_*(n) = \sum_{d|n} G_*(d; n).$$

Its transference identity separates:

- total correlation;
- leakage outside the prime/main-composite regions;
- the exceptional N_2 collar;
- the positive comparison margin.

Balanced R_9 is one generated leakage family that can feed this architecture once the physical source dictionary is proved; it is not the whole fixed-certificate leakage problem.

28.1 Gate 2 consumes, rather than invents, the packet bounds

Gate 2 is the Ford–Maynard lower-bound/endgame compiler. It fixes one compatible ε_* and the shrunk parameter triple

$$P_{\varepsilon_*} = \left(\frac{1}{2} - \varepsilon_*, \varepsilon_*, \frac{1}{6} - 2\varepsilon_* \right).$$

Its source-specific replay requires:

1. Gate-0 comparison regularity, growth and exact Type-I input;
2. the literal $G(d; n)/H(n)$ dictionary and the Ford-generated packet census;
3. the source-specific aggregate N_2 estimate;
4. one compatible fixed ε_* ;
5. either the universal Type-II wrapper, or all generated-(7.23) packet estimates together with **SOURCE-ADAPTER45** and C_{FM} .

Thus the correct closure order is

packet census $\rightarrow$ Gate providers $\rightarrow$ generated Type-II reassembly $\rightarrow$ Gate 2 lower bound,
not Gate 2 followed by a later packet census.

29 Current #287 frontier after Balanced7

The fixed-degree-seven Balanced7 packet is now an audited input to the next source layer, not the first main-line analytic residual. Its status is:

HOSTILE-AUDITED RESEARCH/PAPER PASS
SUBJECT TO EXACT SP-2 SOURCE SEAL

The effective version is separately open.

The next parent is

287-K0-SP2-UNIFORM-FRAGMENTATION-REASSEMBLY45.

Within its determinant-one cross-level sector the latest exact reductions are:

1. short-lift Euler collapse after a uniformity repair;
2. product-modulus compression $\mu(g_1)\mu(g_2) = \mu(r_1r_2)$;
3. fixed- D frequency rigidity and $t_1/g_1 - t_2/g_2 = D/(g_0n)$;
4. CRT reciprocal normal form modulo n ;
5. two-state prime assignment at fixed n , with an explicit firewall against falsely promoting it to multiplicativity in n ;
6. exact shared-gcd Gram-as-square;
7. a provisional coefficient-blind strict-subpolytope energy bound;
8. a negative Ramanujan-completion result showing why the Möbius sign must be used before full primitive-frequency completion.

The controlling one-level object is

$$\mathcal{G}_{H,d} = \sum_{a \succ A} \rho(a) \left| \sum_{\substack{m \succ G/d \\ (m,d)=1}} \frac{\mu(m)}{m} \sum_{t \bmod dm}^* \tilde{G}_{dm,s,\tau}(t) e_{dm}(ta) \right|^2,$$

and the current first exact main-line target is

$$\left| \sum_d \frac{\lambda_H(d)}{d^2} \mathcal{G}_{H,d} \right| \ll AB^2 (\log X)^{-6-2\eta-C_{\text{route}}}.$$

Programme label:

287-K0-SP2-DET1-SHAREDGCD-ONELEVEL-MOBIUS-GRAM45.

This target remains open.

Node	Current status	Publication interpretation
------	----------------	----------------------------

30 Master current-status ledger

Node	Current status	Publication interpretation
Finite exclusion $M \leq 4 \times 10^9$ WindowPairSupply compiler	PUBLIC LEAN-CHECKED PUBLIC LEAN-CHECKED / CONDITIONAL	Strongest unconditional theorem. Global closing compiler; eventual supply open.
Gate 0	MIXED PASS / SOURCE PINS	Physical comparison/Type-I package.
Gate 1A	CONDITIONAL	Use is packet-census dependent.
Gate 1B	CANONICAL PROVIDER OPEN	First exact research residual C4SHIFT-BEZOUT-DUAL-ROELL45; feeding D -census hostile-audit pending and zero-mode local match source-open.
Balanced7 full- q analytic packet	HOSTILE-AUDITED RESEARCH/PAPER PASS	Conditional on exact public SP-2 source seal; not Lean analytic.
Balanced7 source seal	SOURCE-OPEN IN PUBLIC MANUSCRIPT REPO	Later supplied safe bank not synchronized here.
Balanced7 effective	OPEN	Effective low-conductor/exceptional PNT dictionary missing.
287-K0-SP2-UNIFORM- FRAGMENTATION-REASSEMBLY45	OPEN / HISTORICAL R7 PARENT / SUPERSEDED AS CONTROLLING	Must cover actual k -range and all fragmentation/source states.
DET1-LARGESHAREDG0-CELLS45 global closure	RETRACTED	Local fixed-pair gains survive; full cross-level multiplicity was missing.
DET1-PRIMITIVE-NEARFREQ45 global closure	RETRACTED	Near-density gain survives locally but does not close the global family.
DET1-SHORTLIFT-EULER-COLLAPSE45	PASS AFTER UNIFORMITY REPAIR	Analytic zero-free input not Lean formalized.
DET1-LEVELPAIR-PRODUCTMOD-SIGN45	RESEARCH-EXACT PASS / BANK PENDING	Two level signs compress to one $\mu(n)$.
DET1-FIXEDD-FREQUENCY-RIGIDITY45	RESEARCH-EXACT PASS / BANK PENDING	Fixes primitive residues; friable kernel depends on $D/(g_0n)$.
DET1-LEVELPAIR-N- RECIPROCAL-NORMALFORM45	RESEARCH-EXACT PASS / BANK PENDING	CRT reciprocal source modulo n .
DET1-LEVELPAIR-PRIMEASSIGNMENT45	PASS AT FIXED n	Does not imply multiplicativity in n .
DET1-LEVELPAIR-SINGLEMOBIUS-N45	OPEN	Ordinary Euler-product route unavailable.
DET1-SHAREDGCD-GRAM-SQUARE45	RESEARCH-EXACT PASS / BANK PENDING	Compresses cross-level Gram to one-level Möbius operator.
DET1-SHAREDGCD-ONELEVEL-ENERGY45	PROVISIONAL STRICT-SUBPOLYTOPE CLOSURE	Hostile promotion audit pending.
ONELEVEL-RAMANUJAN-ZEROFREE45	NONCLOSING / REPRESENTATION LOOP	Full Ramanujan completion neutralizes Möbius sign.
287-K0-SP2-DET1-SHAREDGCD- ONELEVEL-MOBIUS-GRAM45	FIRST EXACT MAIN-LINE ANALYTIC RESIDUAL / OPEN	Exploit one-level signed Möbius operator before full primitive completion.
FCL	NOT REACHED / OPEN	Downstream of uniform $k = 0$ and remaining fixed-certificate census.
N_2 / Gate 2	NOT ACTIVATED	Downstream.
287-EFFECTIVE-LOWCOND-EXCEPTIONAL- PNT45	OPEN	Effective threshold lane.
Eventual WindowPairSupply	OPEN	Required before finite compiler fires globally.

Node	Current status	Publication interpretation
Erdős Problem #287	OPEN	No unconditional proof claim.

31 Superseded, false and provenance-only routes

The project preserves failed and superseded routes because they explain the source firewalls, but they do not control R7:

- squarefree polarization plus the $X^{6/7+o(1)}$ repeated-prime router: valid historical encoding, superseded by factorial Euler polarization;
- complete multiplicativity as a required repair: retired/not needed;
- the direct Pascadi smooth-number splice at $(3/5, 1/7)$: nonclosing under the audited written parameters;
- V14 Vaughan/prime-modulus two-outer route: valid but nonminimal source audit;
- direct singleton = existing common-conductor Gate-1B covariance: literal source dictionary false;
- direct Gate-1A determinant-one adapter: literal determinant mismatch;
- equal-seven P6.3 three-block route and separated-large-sieve closure: nonclosing;
- DI/Kuznetsov and later HHH-first 3221 lanes: preserved as formal/source infrastructure but superseded after the direct full- q SP-2 audit;
- unique Ford-(7.23) Balanced7 packet interpretation: retracted; the literal source is fixed-certificate SP-2;
- SmallQ Gate-0 Type-I: wrong provider for $q \mid 2P + s$, not a false theorem in its proper source class;
- large-shared- g_0 and primitive-near-frequency *global closure*: retracted after restoring the G^2/H cross-level multiplicity and full normalizations; their local fixed-pair cancellation remains valid;
- PRIMITIVE-SMALLGCD-FAR-HARDDEN-GRAM45: proposed but never promoted as the controlling residual;
- ordinary Euler-product cancellation from the fixed- n two-state assignment: unavailable because the local factor depends on the complementary coprime component;
- complete primitive- t Ramanujan zero-free route: nonclosing because its local sign generically neutralizes the Möbius sign.

32 Formal theorem and provenance map

Public Lean bank

The public manuscript repository contains the exact problem predicate, finite exclusion through 4×10^9 , WindowPairSupply compiler, fixed-certificate source compilers, V15–V19 affine/factorial/3221 modules and their aggregate imports.

Supplied later safe bank

A later supplied report records an 8213-job successful append-only build with seven-box weights, product multiplicity, conductor-split large-sieve compiler, low-conductor Siegel–Walfisz interface, SmallQ compiler, small-prime-prefix source, post-repair owner compiler, uniform-fragmentation compiler and post-Balanced7 status audit. Those modules are not currently present in this public manuscript repository. Their R7 status is SUPPLIED LEAN SAFE BANK / REPOSITORY SYNC PENDING.

Already supplied determinant-one formal bank

The supplied formal work already banks finite primitive Ramanujan algebra/reassembly, short-lift finite profiles, harmonic twists, shared- g_0 u -parameter routing, averaged b -pair gcd identities, reduced-denominator algebra and exact near-frequency counting. No deep analytic input is thereby inhabited.

Latest research-exact layer, formal banking pending

The product-modulus Möbius-sign compression, fixed- D frequency rigidity, CRT n -reciprocal normal form and shared-gcd Gram-as-square are currently research-exact mathematics. R7 does not call them Lean-verified until a corresponding successful formal bank is supplied and synchronized.

External analytic boundary

The Balanced7 hostile audit uses external multiplicative large-sieve, Siegel–Walfisz, one-dimensional upper-sieve and Shiu inputs. The short-lift uniformity repair uses a zero-free-region analytic statement. The first one-level Möbius Gram remains open. Thus the formal compiler/source bank and the analytic research status remain deliberately separate.

References

- [1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L’Enseignement Mathématique, 1980, p. 33.
- [2] T. F. Bloom, *Erdős Problem #287*, Erdős Problems archive, problem provenance and status page, accessed August 2026.
- [3] K. Ford and J. Maynard, *On the theory of prime producing sieves*, arXiv:2407.14368, 2024.
- [4] S. Drappeau, A. Granville and X. Shao, *Smooth-supported multiplicative functions in arithmetic progressions beyond the $x^{1/2}$ -barrier*, Mathematika 63 (2017), 895–918.
- [5] A. Pascadi, *On the exponents of distribution of primes and smooth numbers*, arXiv:2505.00653v2, 29 June 2025.
- [6] The Lean and Mathlib communities, *Lean 4 and Mathlib*, formal proof assistant and mathematical library.

31 August 2026 historical final status (superseded by R9)

Finite $M \leq 4 \times 10^9$	KERNEL-PROVED / unchanged
C0 exact-product / Double Type II	ANALYTICALLY BANKED
C0	ANALYTICALLY CLOSED / CONDITIONAL SOURCE PIN
Formal source pin	SHAREDGCD-OMEGAH-NORMALISATION-SOURCEPIN45
Full transverse branch	STRICTLY REDUCED / OPEN
First transverse residual	CRITICAL-BEZOUTNUMERATOR-SINGLECARRIER45
Parallel b -diagonal	OPEN
Supplied latest formal build	8298 jobs / 0 errors; public dependency sync pending
Erdős #287	OPEN

The historical Gate 0–2 theory, R7 uniform- $k = 0$ branch, Ford/FCL architecture, and finite compiler are preserved above. No implication from the local C0/transverse milestones to a complete solution is asserted. Downstream Twin Prime work, if referenced, is a separate programme and is not proved by this manuscript.